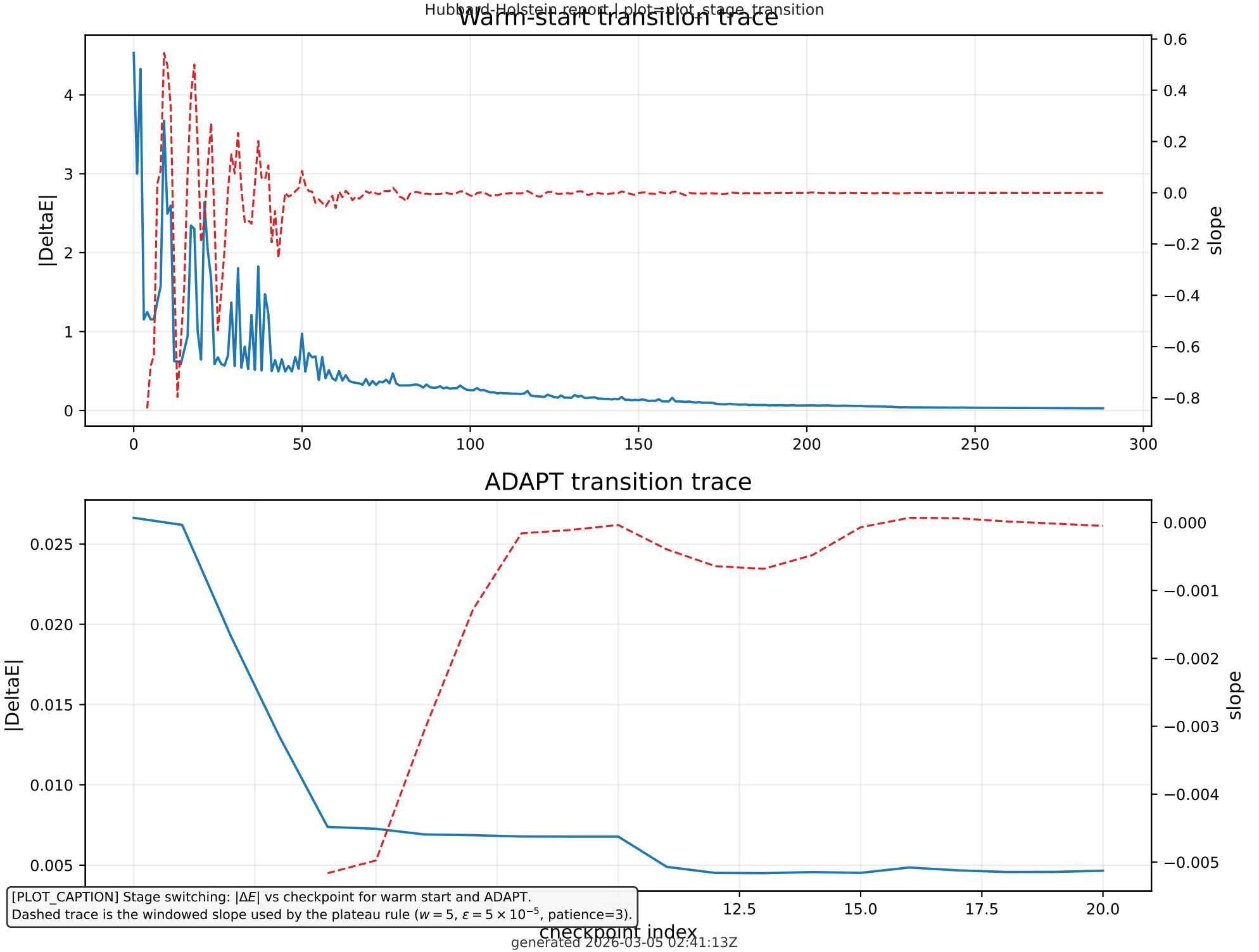
```
============================================================
  PARAMETER MANIFEST
============================================================

  Model           : Hubbard-Holstein
  Ansatz          : warm:hh_hva_ptw -> adapt:PoolB(UCCSD+HVA+PAOP_FULL) -> final:hh_hva_ptw
  Drive enabled   : True

  t (hopping)     : 1.0
  U (interaction) : 4.0
  dv (disorder)   : 0.0

  L               : 2
  n_ph_max        : 1
  omega0          : 1.0
  g_ep            : 0.5
  noise_modes     : ['shots', 'aer_noise']
  noisy_methods   : ['suzuki2', 'cfqm4']
  trotter_steps   : 128
  num_times       : 201
  magnus_variants : midpoint_magnus2,cfqm4,cfqm6
  hardcoded_family: included(static+drive)

  Command:
    /opt/anaconda3/bin/python /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/pipelines/exa
    ct_bench/hh_noise_robustness_seq_report.py --L 2 --t 1.0 --u 4.0 --dv 0.0 --omega0 1.0 --g-ep 0.5 --n-ph-
    max 1 --boundary open --ordering blocked --warm-reps 2 --warm-restarts 4 --warm-maxiter 2400 --warm-method
    COBYLA --warm-seed 7 --adapt-max-depth 80 --adapt-maxiter 3000 --adapt-eps-grad 5e-7 --adapt-eps-energy
    1e-9 --adapt-seed 7 --adapt-allow-repeats --final-reps 3 --final-restarts 4 --final-maxiter 2400 --final-
    method COBYLA --final-seed 7 --t-final 15.0 --num-times 201 --trotter-steps 128 --exact-steps-multiplier 2
    --include-drive-profile --drive-A 0.5 --drive-omega 2.0 --drive-tbar 3.0 --drive-phi 0.0 --drive-pattern
    staggered --drive-time-sampling midpoint --noise-modes shots,aer_noise --noisy-methods suzuki2,cfqm4
    --shots 4096 --oracle-repeats 6 --oracle-aggregate mean --noise-seed 7 --use-fake-backend --backend-name
    FakeManilaV2 --no-allow-aer-fallback --omp-shm-workaround --noisy-mode-timeout-s 21600 --output-json /User
```

```
    s/jakestrobel/Documents/Holstein_implementation/Holstein_test/artifacts/json/hh_noise_robustness_L2_fix1B_
    heavy_phase2d_all_noisy_strict.json --output-pdf /Users/jakestrobel/Documents/Holstein_implementation/Hols
    tein_test/docs/HH_noise_robustness_L2_fix1B_heavy_phase2d_all_noisy_strict.pdf

============================================================
```

```
SECTION: RESULTS SUMMARY

HH Noise Robustness Report Summary

final delta_abs: 0.019166429636473498
warm stop: slope_plateau
adapt stop: slope_plateau
final stop: slope_plateau
noisy modes completed: 1 / 4
noisy method-modes completed: 1 / 8

Transition policies:
  warm->adapt: {'window_k': 5, 'slope_epsilon': 5e-05, 'patience': 3, 'min_points_before_switch': 8}
  adapt->vqe: {'window_k': 5, 'slope_epsilon': 5e-05, 'patience': 3, 'min_points_before_switch': 8}

Definitions:
  DeltaE_k = E_k - E_exact_sector
  switch when abs(slope( |DeltaE| window )) <= epsilon for patience checkpoints

Noisy-vs-noiseless policy:
  noisy methods: ['suzuki2', 'cfqm4'] with modes ['shots', 'aer_noise'].
  Magnus/CFQM remain in noiseless matrix; noisy matrix includes selected methods above.

SECTION: HARDCODED FAMILY REMAP
  legacy lane -> warm-start HVA seed
  paop lane   -> ADAPT Pool-B seed
  hva lane    -> final seeded conventional VQE
```

SECTION: RESULTS STAGE TRANSITIONS

Transition traces and Pool-B composition.



[PLOT_CAPTION] Stage switching: $|\Delta E|$ vs checkpoint for warm start and ADAPT.
Dashed trace is the windowed slope used by the plateau rule ($w = 5$, $\varepsilon = 5 \times 10^{-5}$, patience=3).

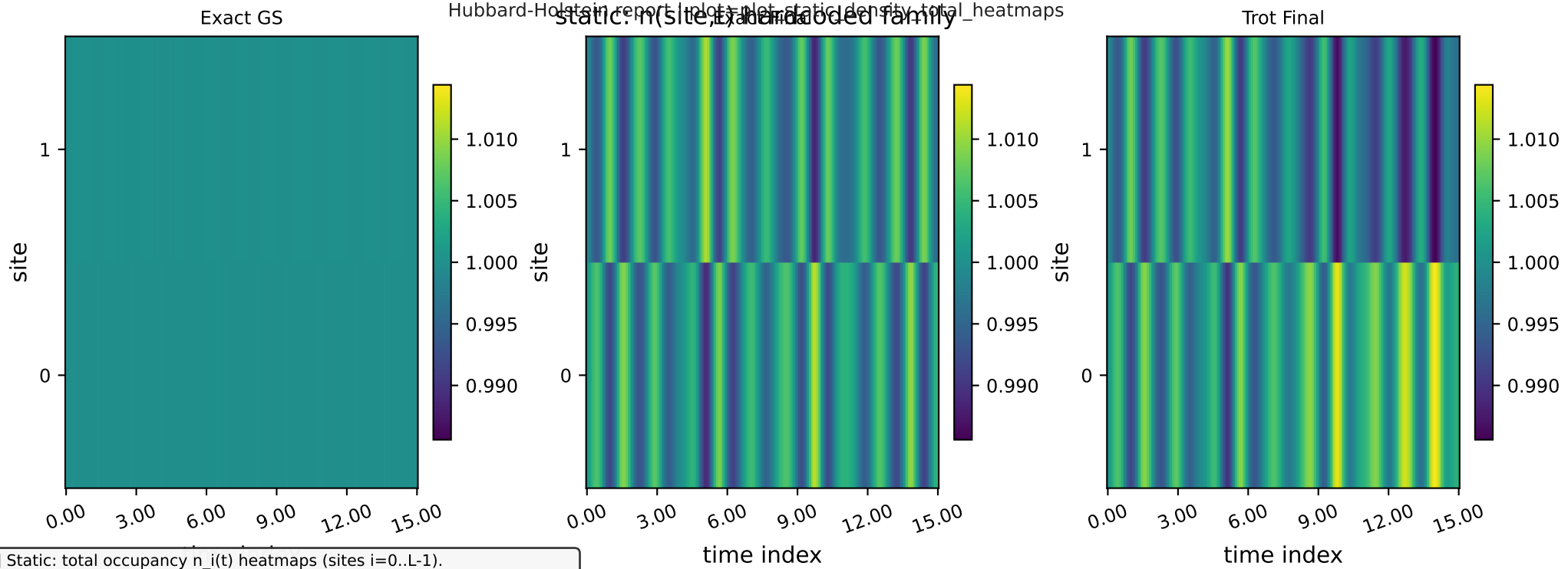
Pool B Composition (strict union + dedup)

Metric	Value
raw uccsd	3
raw hva	8
raw paop_full	9
dedup total	20
presence uccsd	3
presence hva	8
presence paop_full	9
overlap count	0

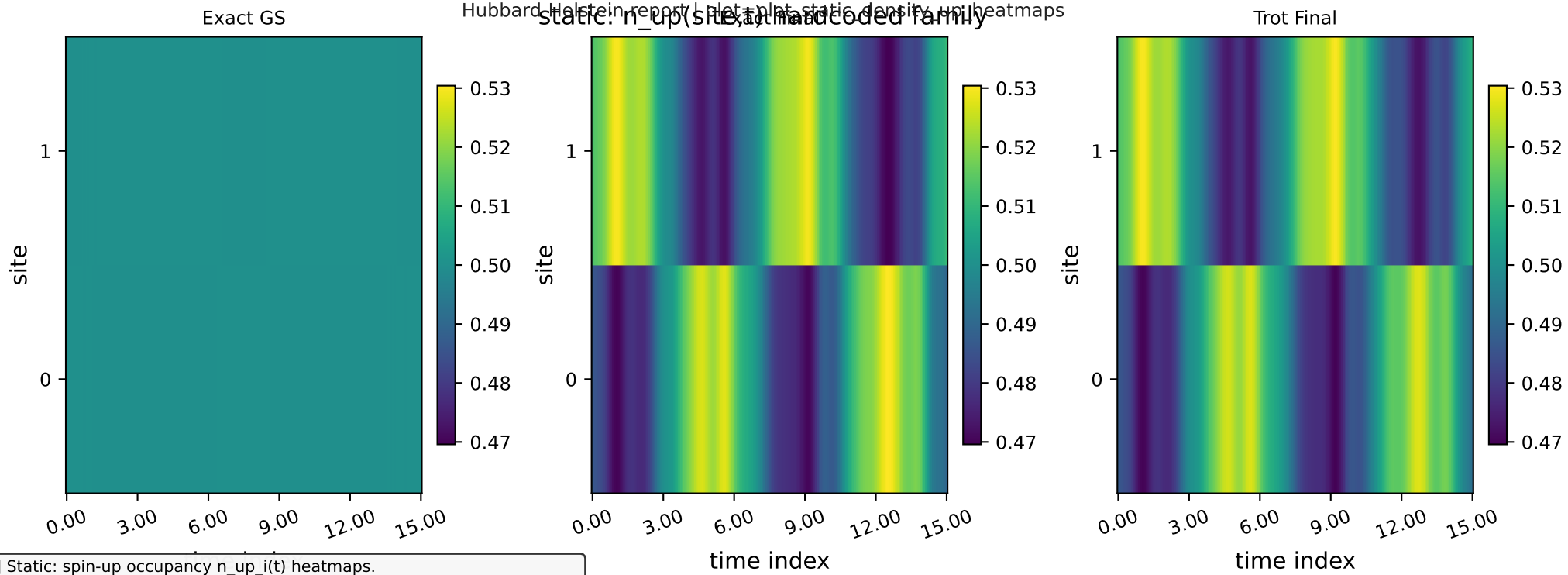
[PLOT_CAPTION] Stage switching: $|\Delta E|$ vs checkpoint for warm start and ADAPT.
Dashed trace is the windowed slope used by the plateau rule ($w = 5$, $\varepsilon = 5 \times 10^{-5}$, patience=3).

SECTION: RESULTS STATIC OVERLAYS

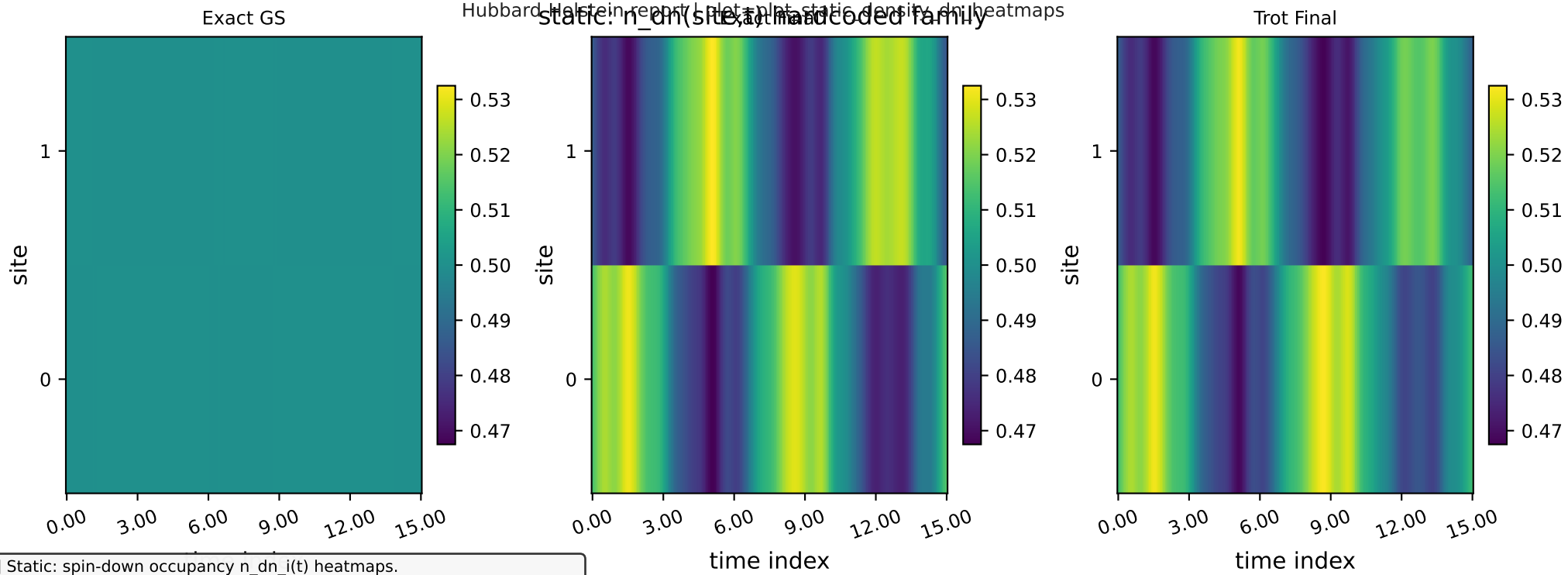
Static-profile overlays and hardcoded-family diagnostics.



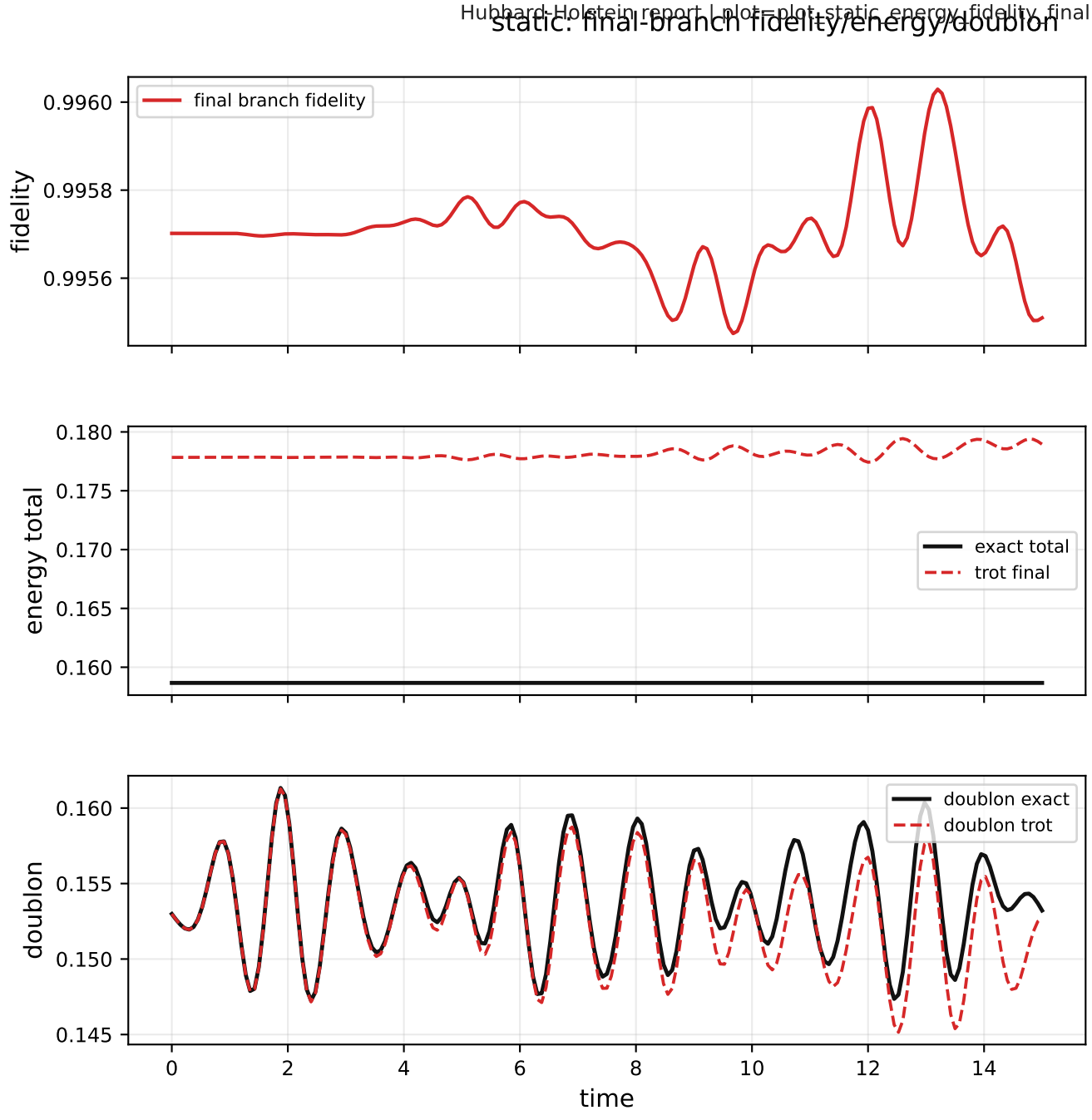
[PLOT_CAPTION] Static: total occupancy n_i(t) heatmaps (sites i=0..L-1).
Panels: exact GS (left), exact evolution (middle), noiseless ADAPT-HVA + Suzuki-2 (right).



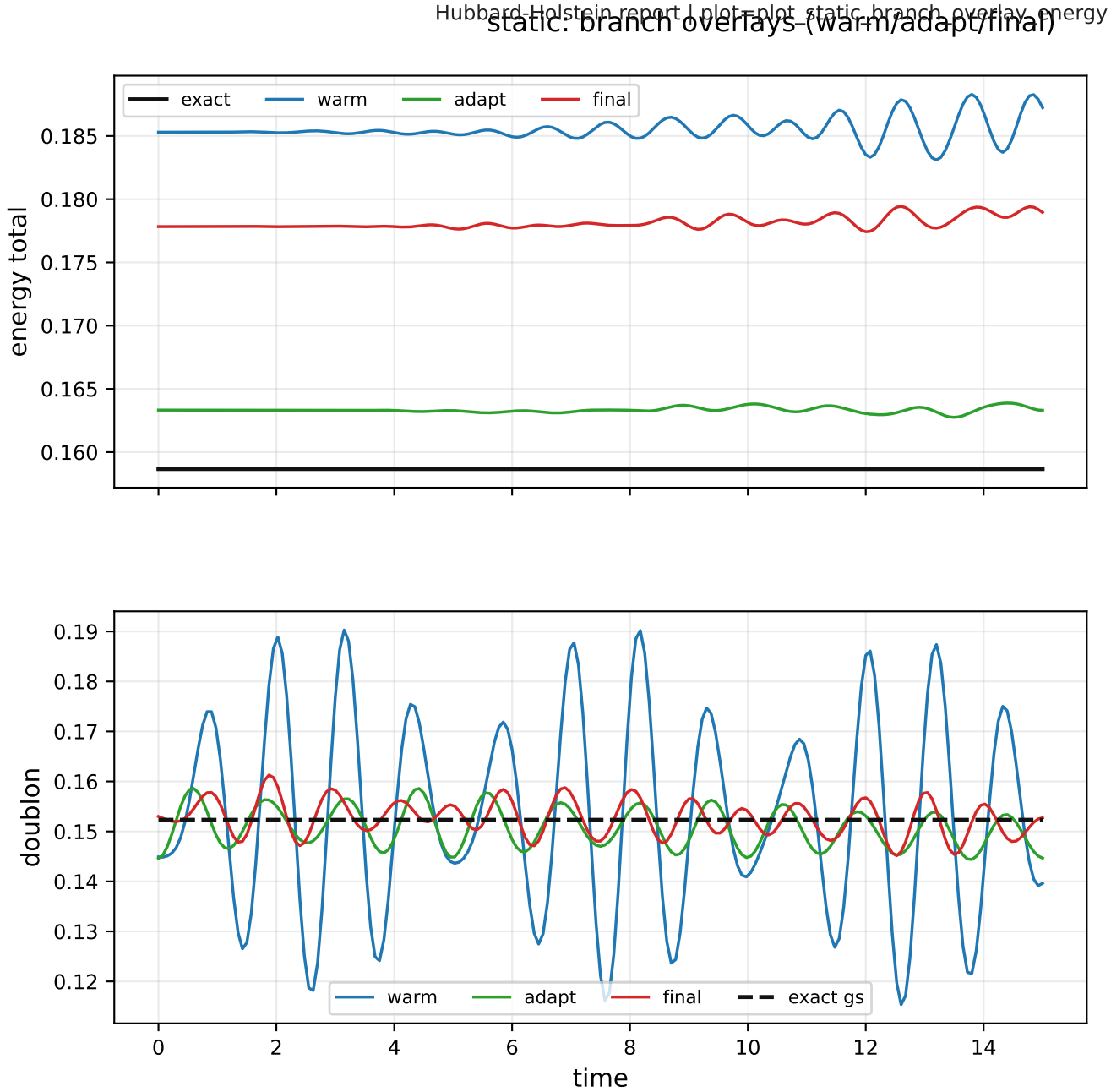
[PLOT_CAPTION] Static: spin-up occupancy n_up_i(t) heatmaps.
Panels: exact GS (left), exact evolution (middle), noiseless ADAPT-HVA + Suzuki-2 (right).



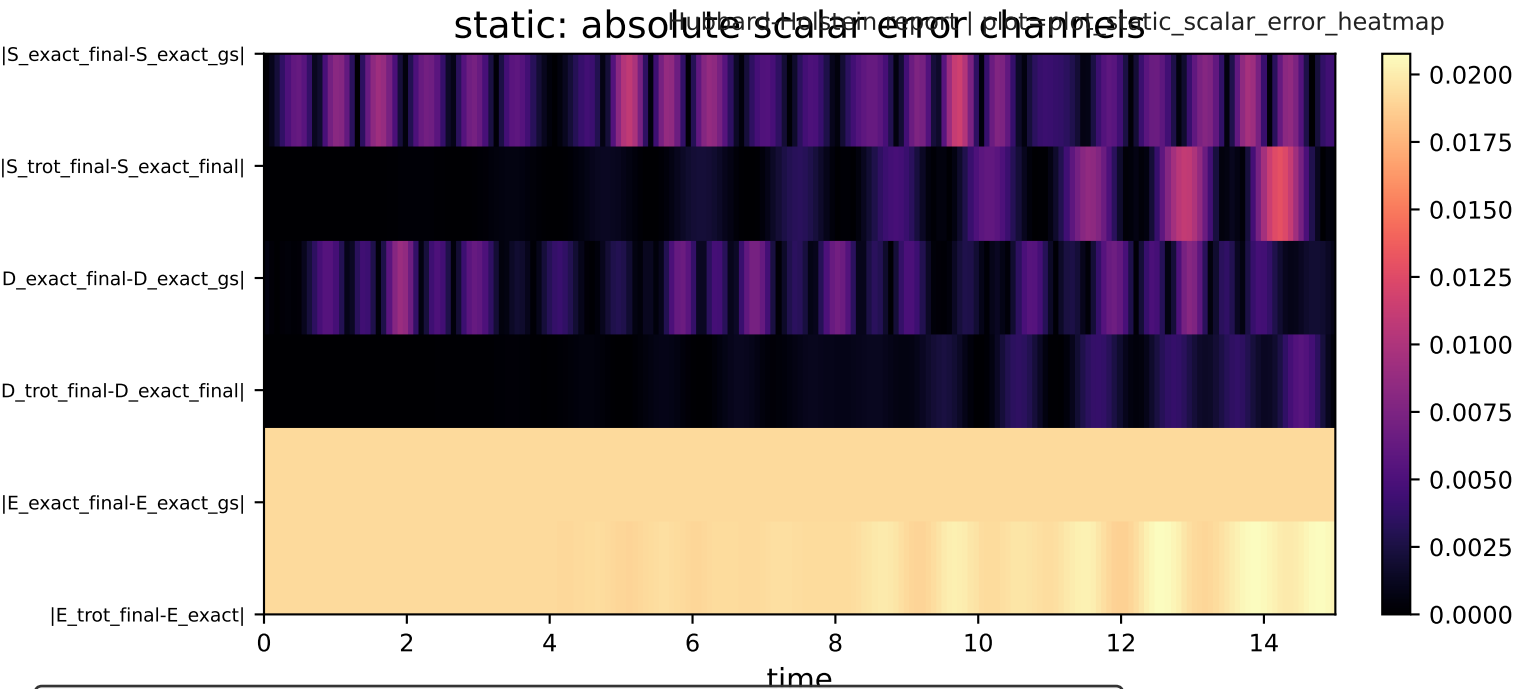
[PLOT_CAPTION] Static: spin-down occupancy n_dn_i(t) heatmaps.
Panels: exact GS (left), exact evolution (middle), noiseless ADAPT-HVA + Suzuki-2 (right).



[PLOT_CAPTION] Static (final branch): top F_sub, middle E_total, bottom doublon vs time.
Exact reference vs noiseless ADAPT-HVA + Suzuki-2.



[PLOT_CAPTION] Static: branch overlays - top E_total, bottom doublon (warm / ADAPT / final).
All curves are noiseless; exact reference shown for comparison.



[PLOT_CAPTION] Static: absolute scalar error channels heatmap (rows are |E|, |D|, and |S| error metrics). Columns run over time; brighter color means larger deviation from the matched exact channel.

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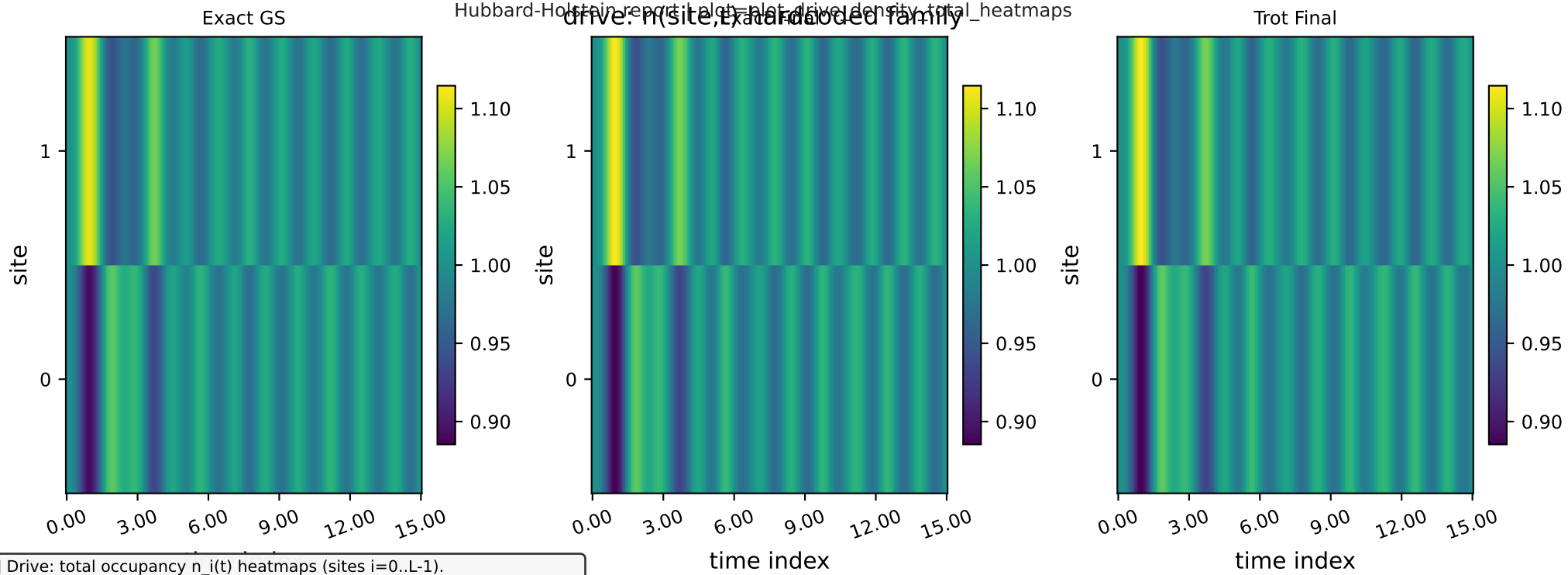
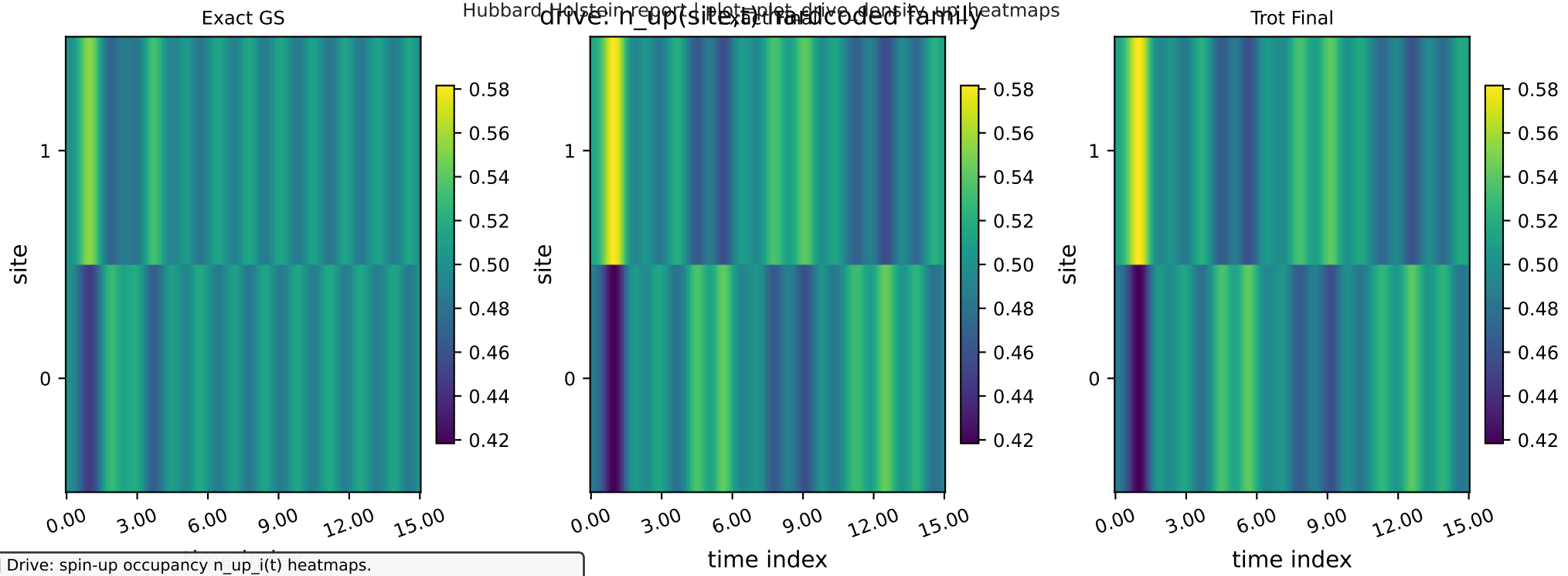
SECTION: RESULTS DRIVE OVERLAYS

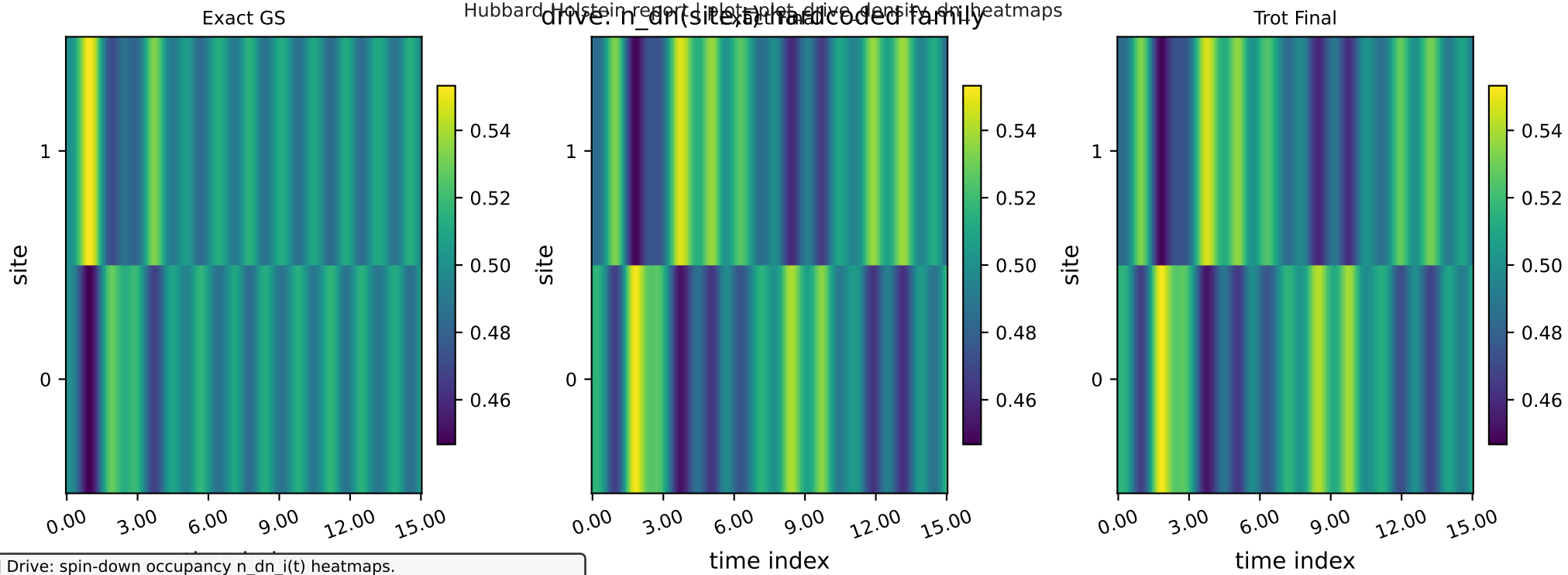
Drive-profile overlays and hardcoded-family diagnostics.



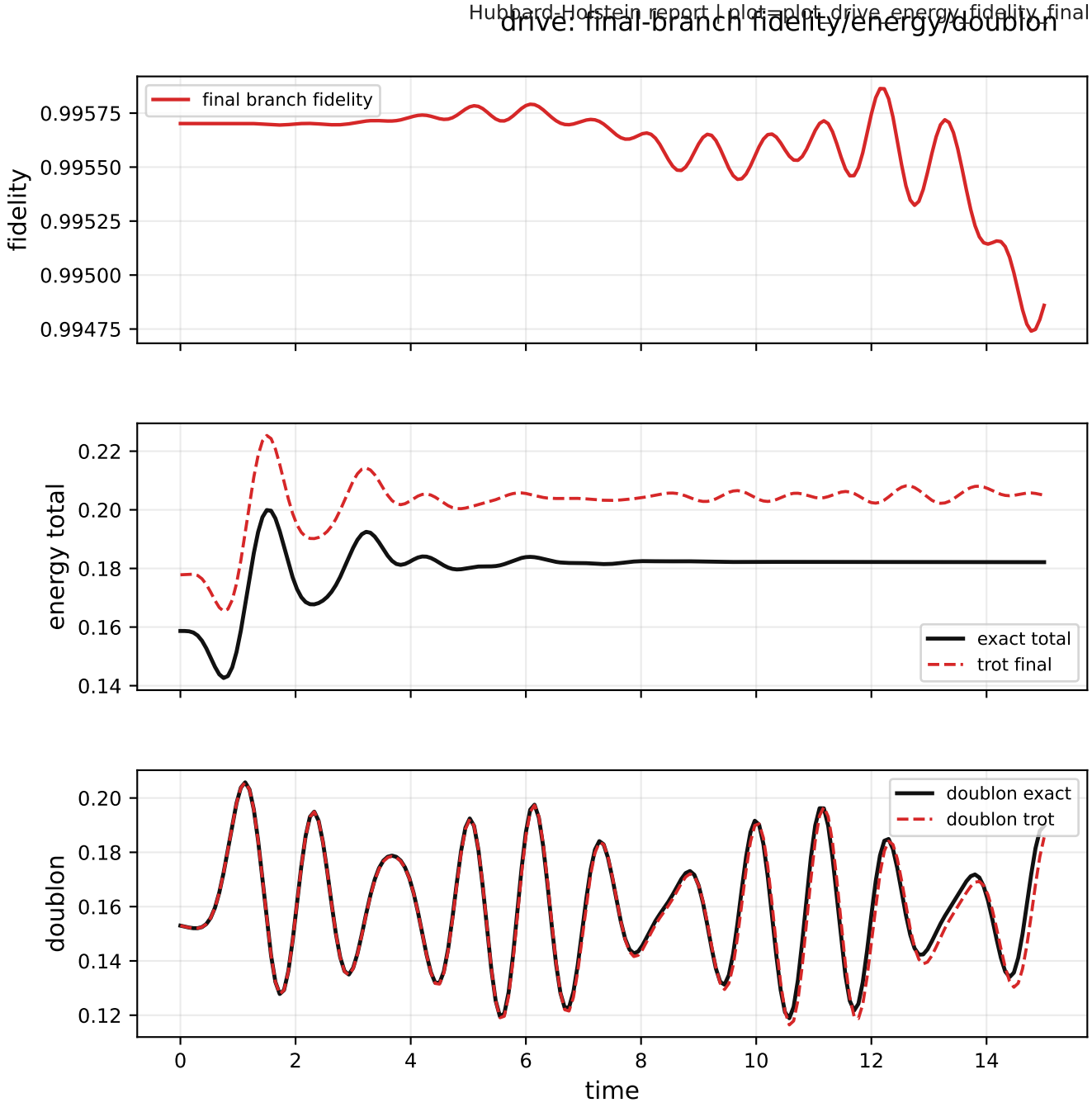
[PLOT_CAPTION] Drive: total occupancy n_i(t) heatmaps (sites i=0..L-1).
Panels: exact GS (left), exact evolution (middle), noiseless ADAPT-HVA + Suzuki-2 (right).



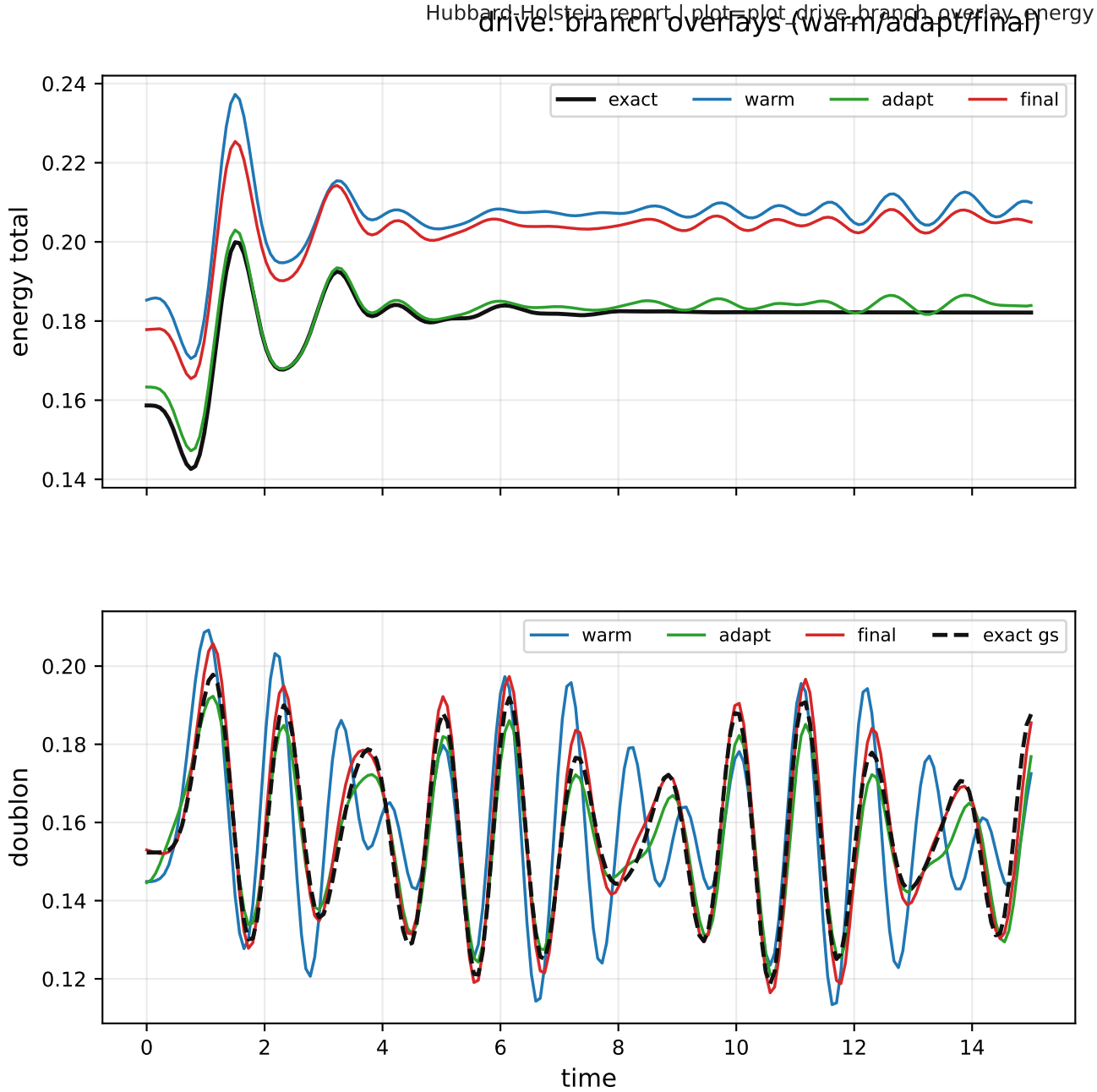
[PLOT_CAPTION] Drive: spin-up occupancy n_up_i(t) heatmaps.
Panels: exact GS (left), exact evolution (middle), noiseless ADAPT-HVA + Suzuki-2 (right).



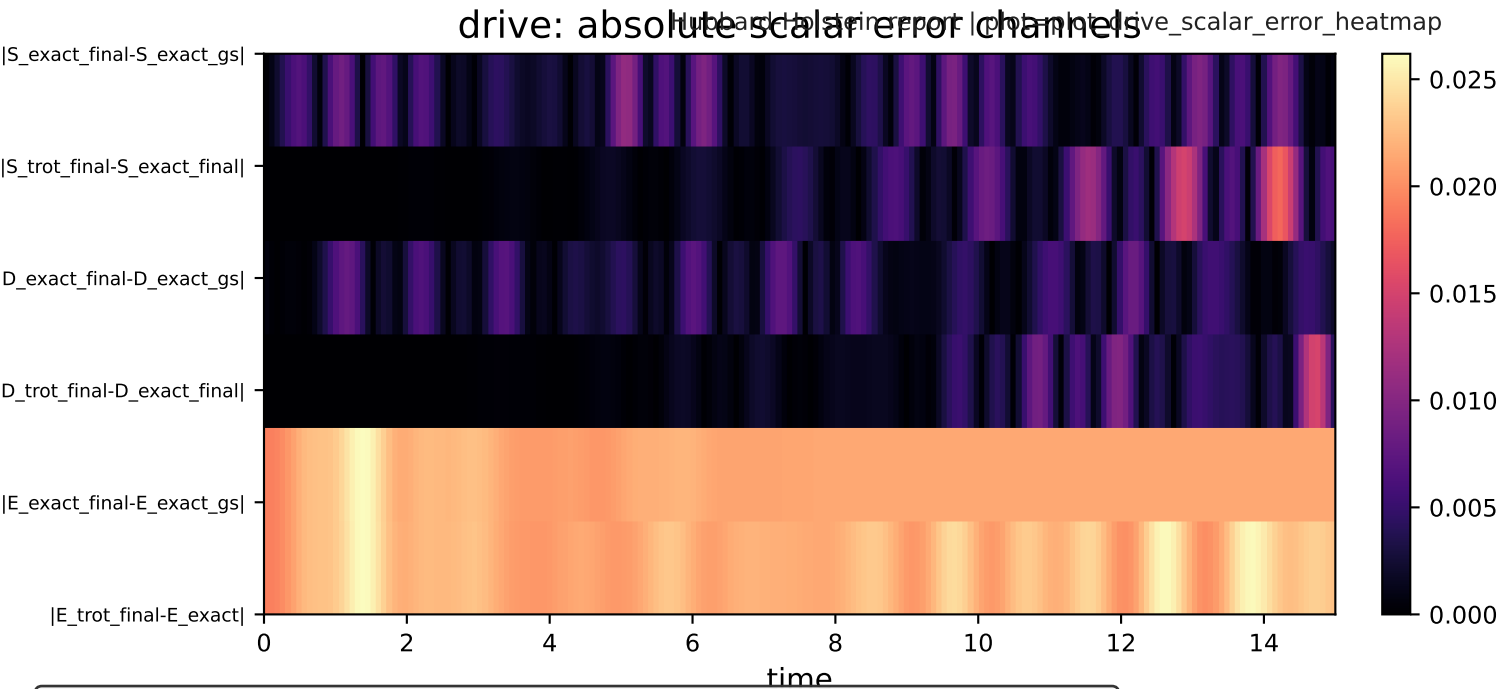
[PLOT_CAPTION] Drive: spin-down occupancy n_dn_i(t) heatmaps.
Panels: exact GS (left), exact evolution (middle), noiseless ADAPT-HVA + Suzuki-2 (right).



[PLOT_CAPTION] Drive (final branch): top F_sub, middle E_total, bottom doublon vs time.
Exact reference vs noiseless ADAPT-HVA + Suzuki-2.



[PLOT_CAPTION] Drive: branch overlays - top E_total, bottom doublon (warm / ADAPT / final).
All curves are noiseless; exact reference shown for comparison.



[PLOT_CAPTION] Drive: absolute scalar error channels heatmap (rows are |E|, |D|, and |S| error metrics). Columns run over time; brighter color means larger deviation from the matched exact channel.

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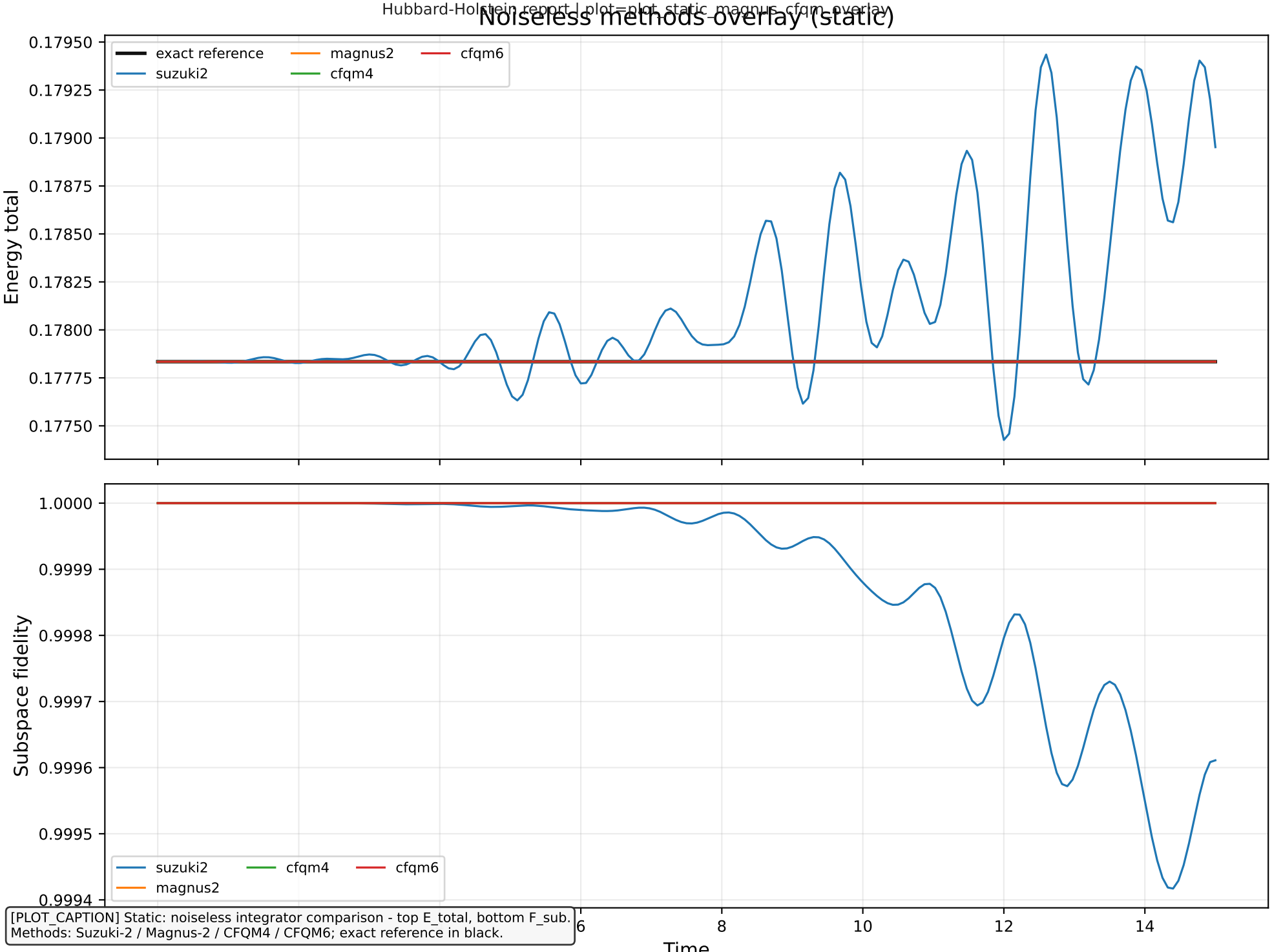
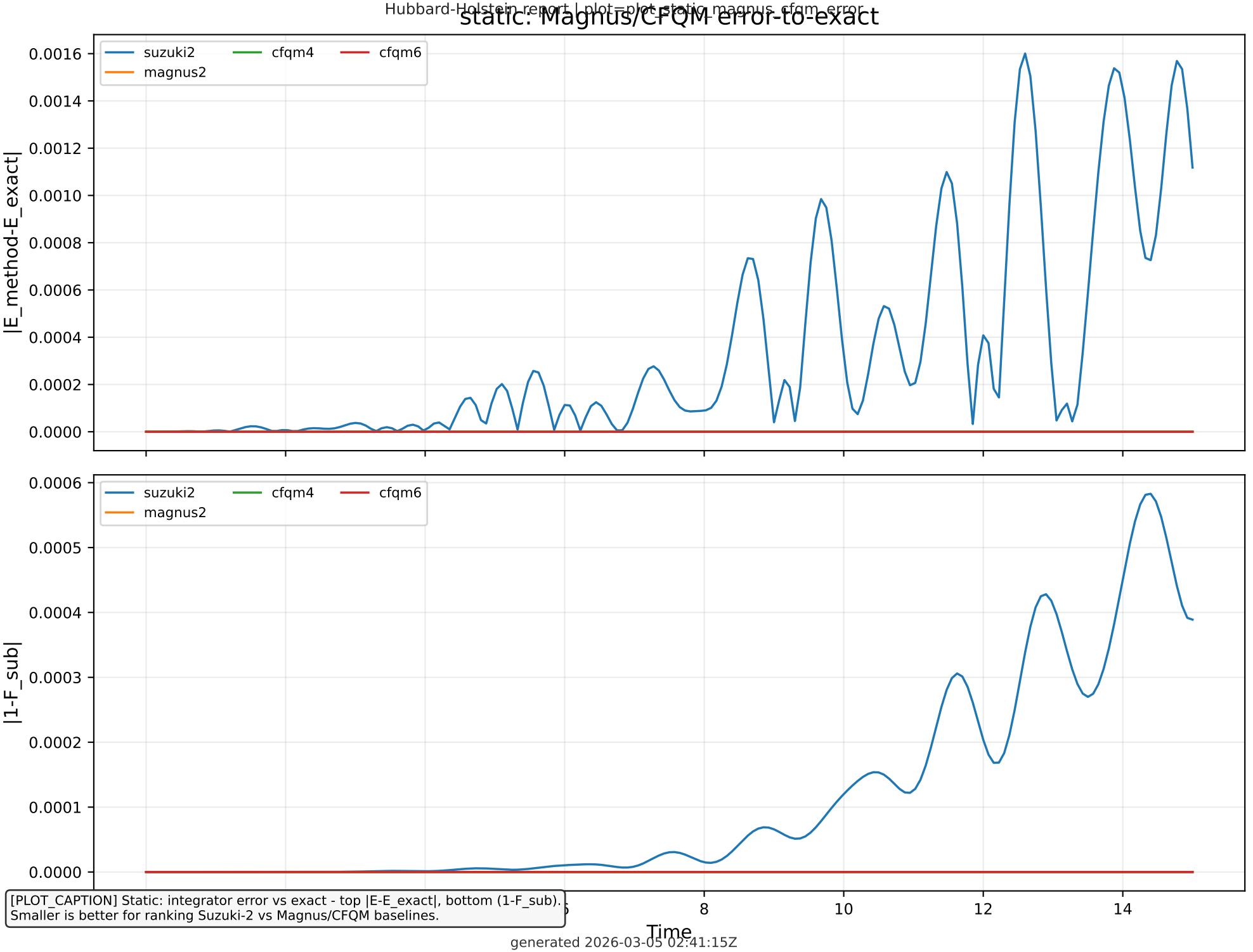
SECTION: RESULTS METHOD COMPARISON

Suzuki / Magnus / CFQM comparison overlays versus exact reference.

Legend key (applies to this section):
  Exact: exact reference evolution.
  Noiseless: ADAPT-HVA pipeline on noiseless simulator.
  Integrators: Suzuki-2, Magnus-2, CFQM4, CFQM6.

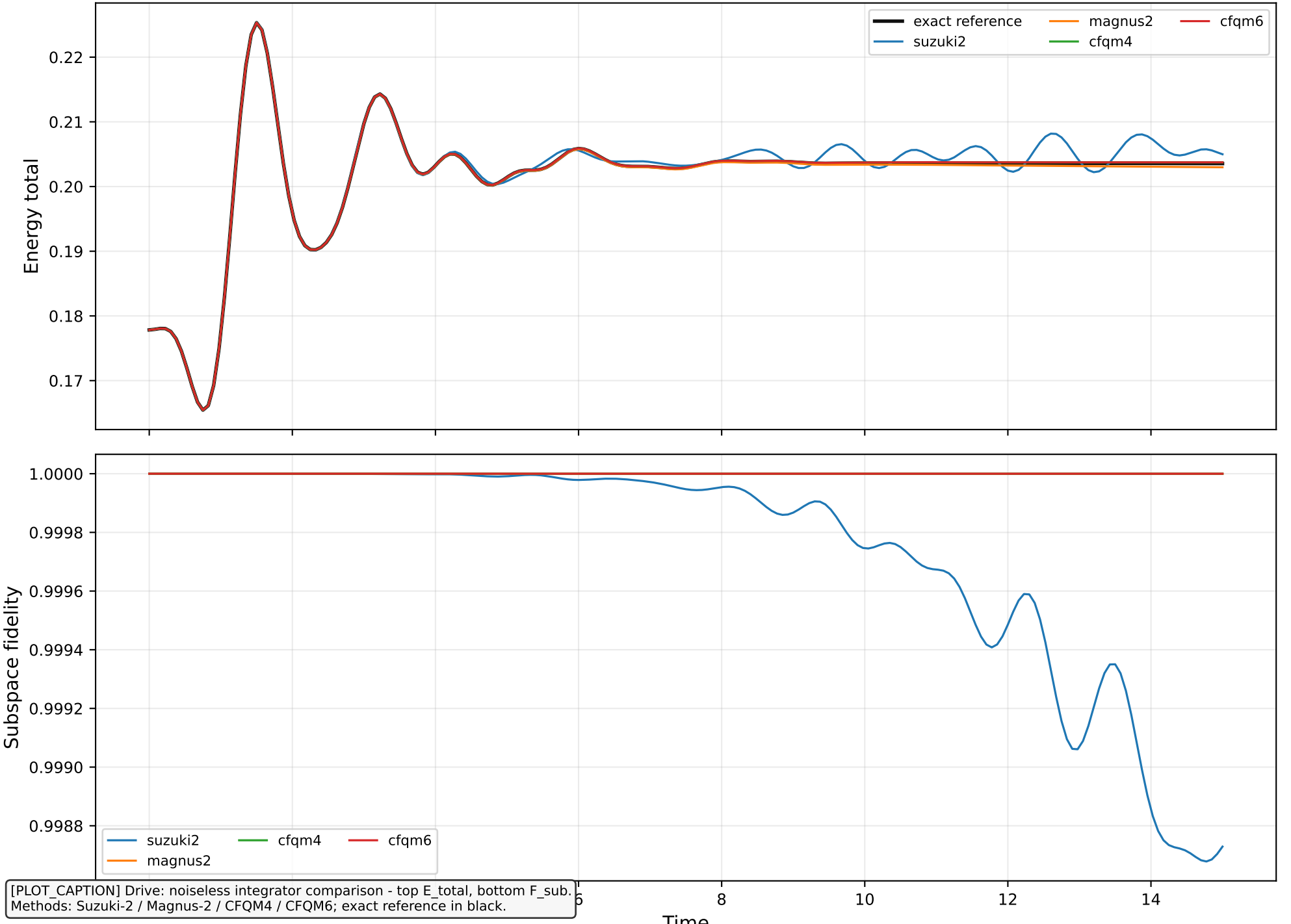


[PLOT_CAPTION] Static: noiseless integrator comparison - top E_total, bottom F_sub.
Methods: Suzuki-2 / Magnus-2 / CFQM4 / CFQM6; exact reference in black.

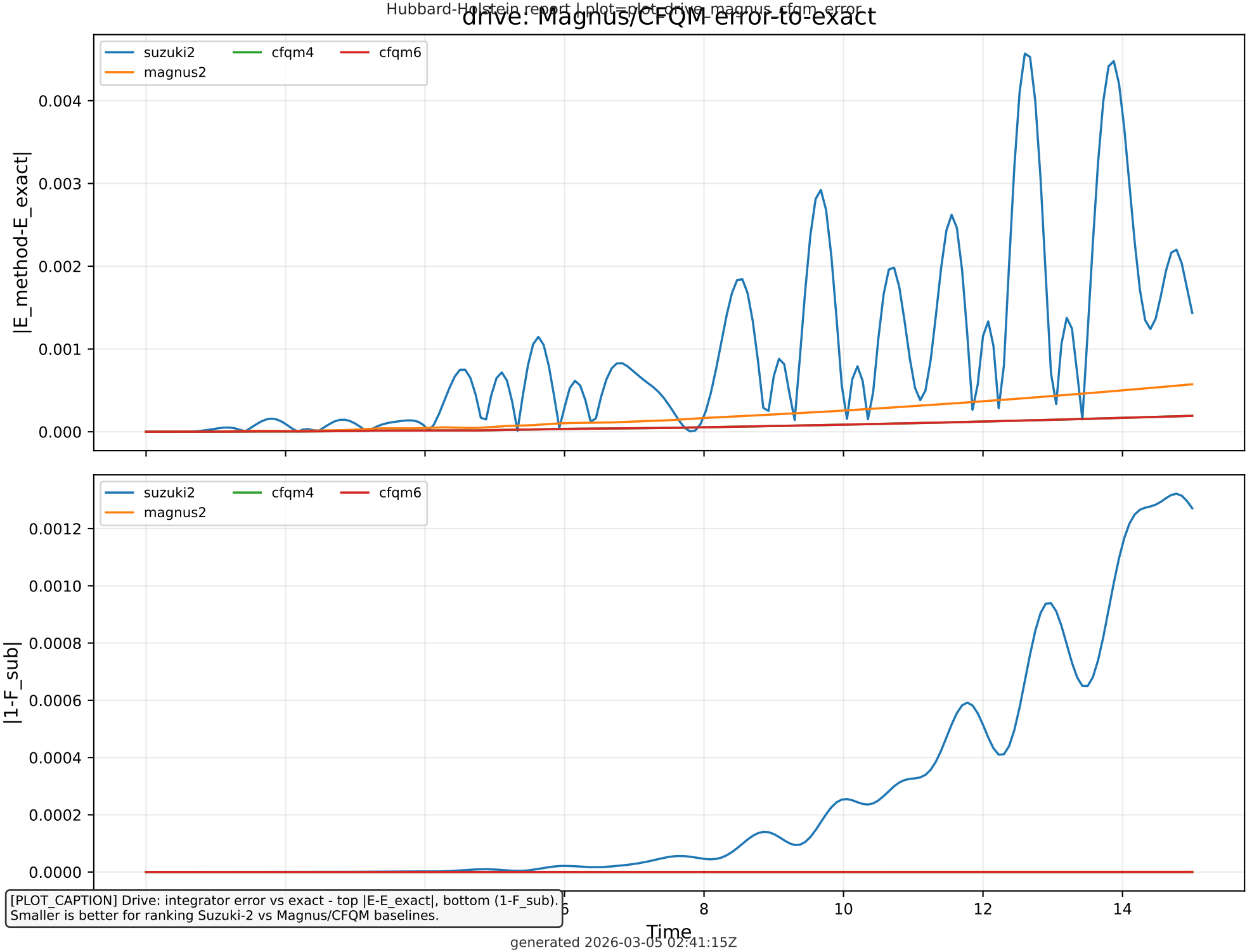


[PLOT_CAPTION] Static: integrator error vs exact - top |E-E_exact|, bottom (1-F_sub).
Smaller is better for ranking Suzuki-2 vs Magnus/CFQM baselines.

Noiseless methods overlay (drive)



[PLOT_CAPTION] Drive: noiseless integrator comparison - top E_total, bottom F_sub.
Methods: Suzuki-2 / Magnus-2 / CFQM4 / CFQM6; exact reference in black.

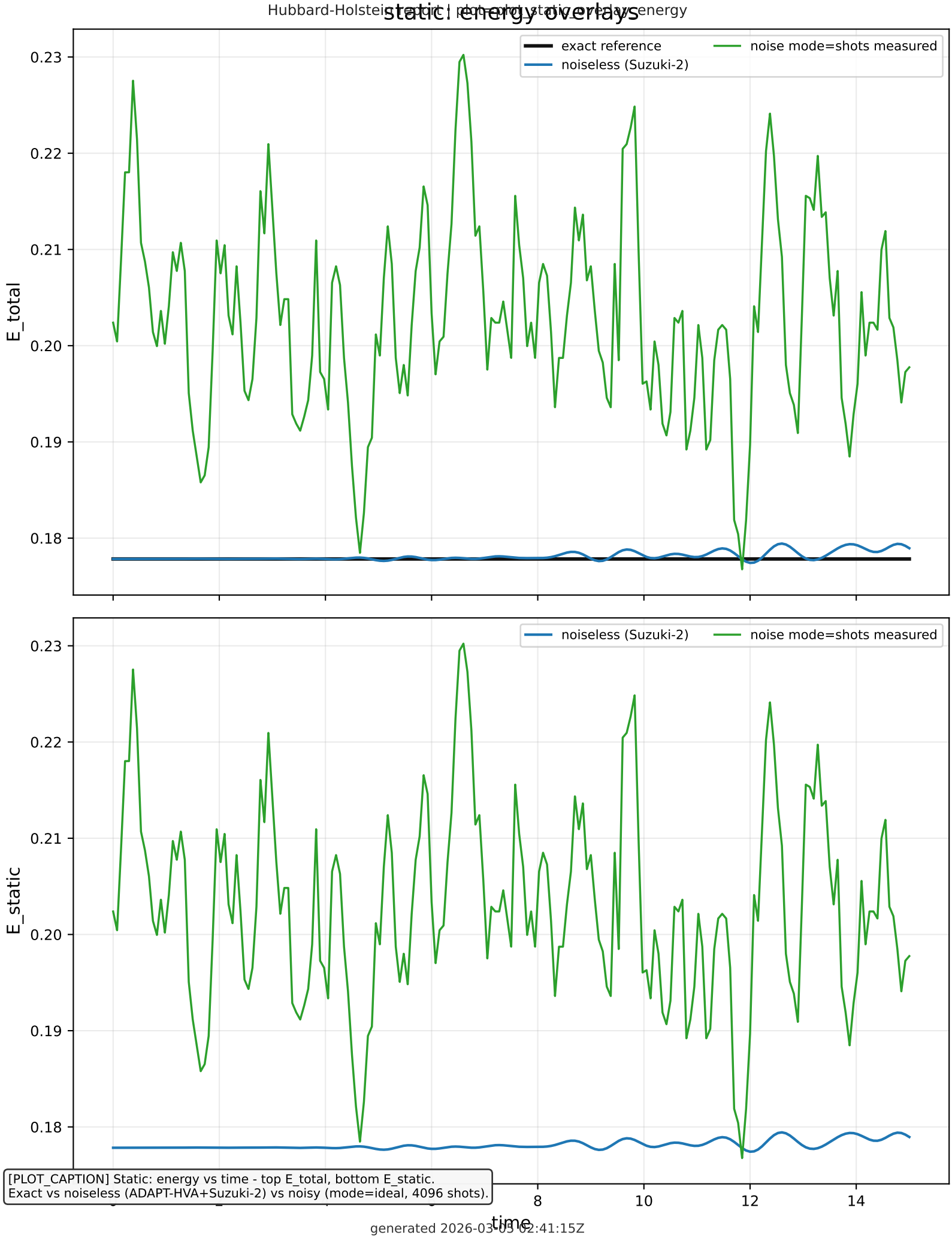


[PLOT_CAPTION] Drive: integrator error vs exact - top |E-E_exact|, bottom (1-F_sub).
Smaller is better for ranking Suzuki-2 vs Magnus/CFQM baselines.

SECTION: RESULTS NOISE DETAILS

Overlay + per-mode detail pages for noise robustness.

Legend key (applies to this section):
  Exact: exact reference evolution.
  Noiseless: ADAPT-HVA pipeline on noiseless simulator.
  Noisy: shot-based estimator under selected mode (4096 shots).
  Ideal-reference: baseline used for Delta(noisy-ideal) panels.


Hubbard-Holstein static | noise=shots static_replay_energy
static: energy overlays
exact reference
noiseless (Suzuki-2)
noise mode=shots measured
E_total
0.23
0.22
0.21
0.20
0.19
0.18
noiseless (Suzuki-2)
noise mode=shots measured
E_static
0.23
0.22
0.21
0.20
0.19
0.18
8
10
12
14
time
[PLOT_CAPTION] Static: energy vs time - top E_total, bottom E_static.
Exact vs noiseless (ADAPT-HVA+Suzuki-2) vs noisy (mode=ideal, 4096 shots).
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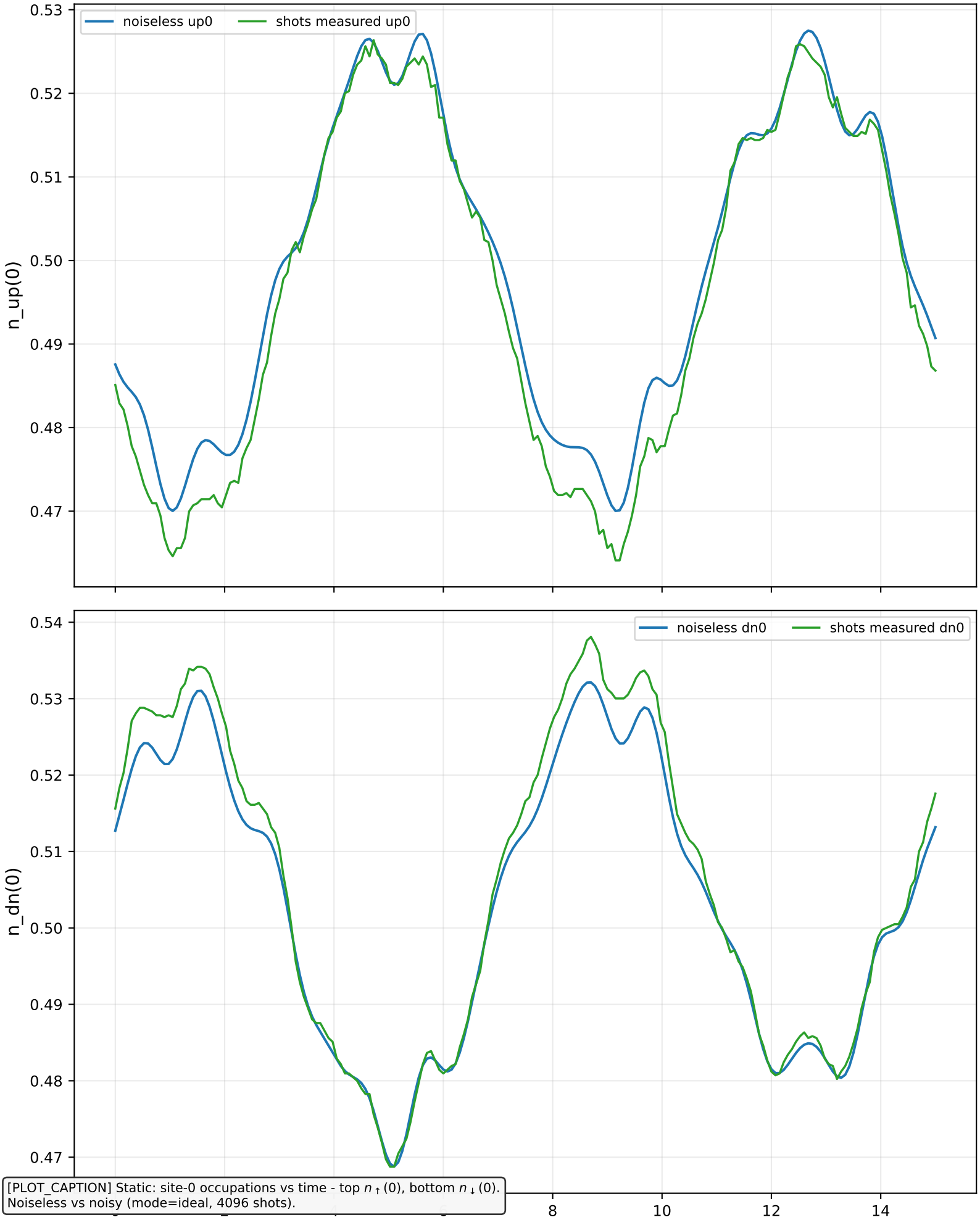
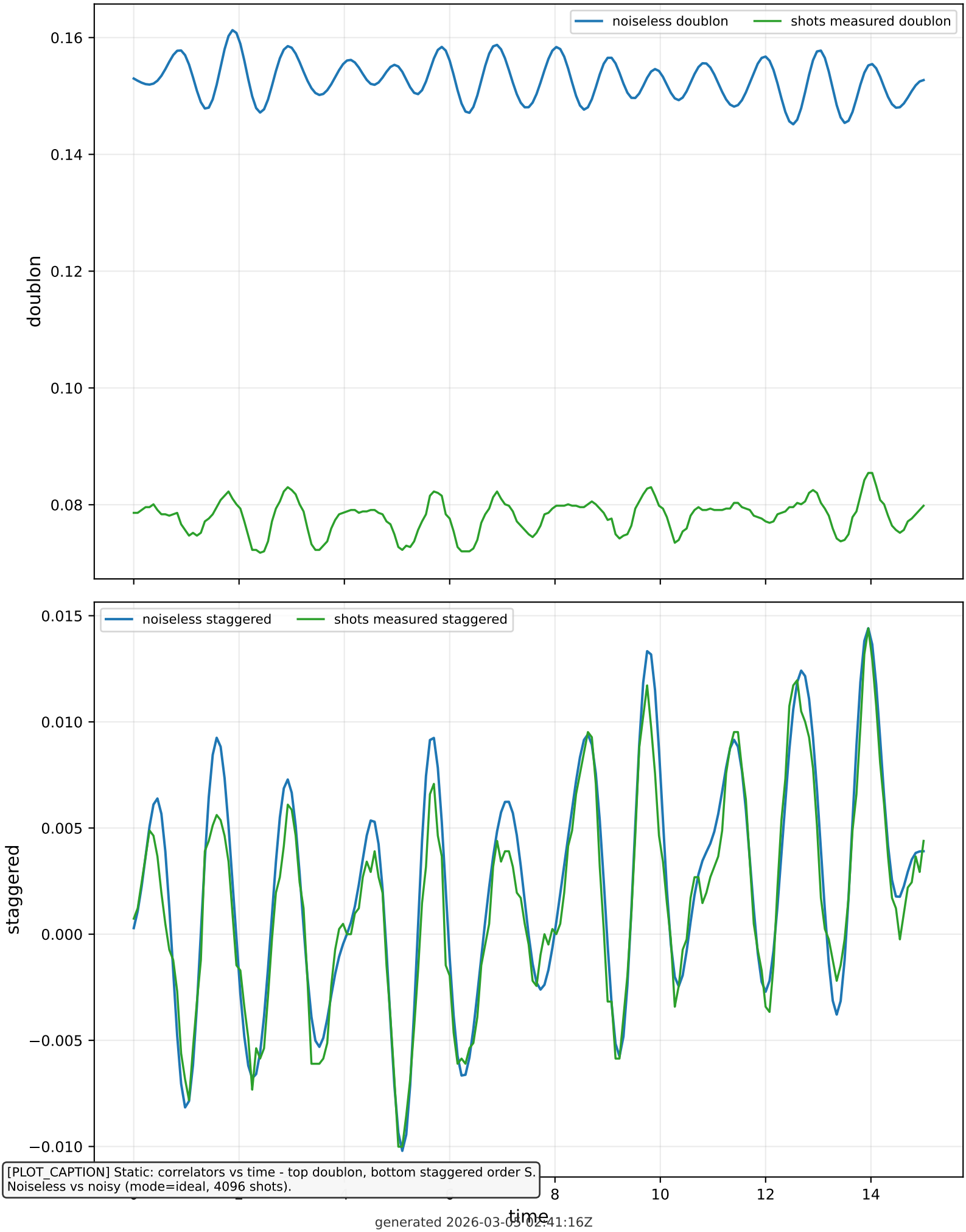
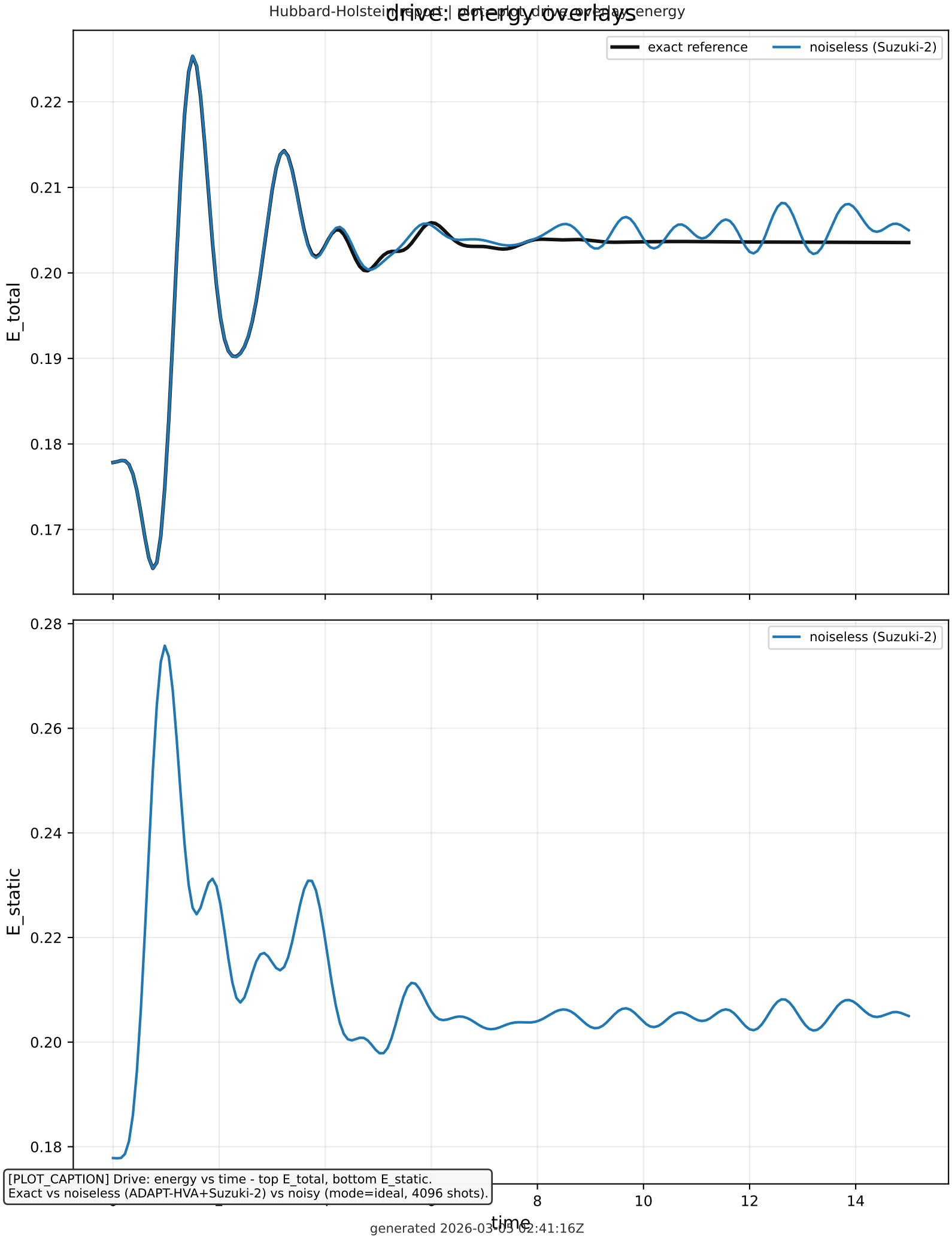
Static: site-0 occupation overlays



[PLOT_CAPTION] Static: site-0 occupations vs time - top $n_\uparrow(0)$, bottom $n_\downarrow(0)$. Noiseless vs noisy (mode=ideal, 4096 shots).



[PLOT_CAPTION] Static: correlators vs time - top doublon, bottom staggered order S.
Noiseless vs noisy (mode=ideal, 4096 shots).


Hubbard-Holstein dimer | adaptive drive overlays
drive: energy overlays
exact reference
noiseless (Suzuki-2)
E_total
0.22
0.21
0.20
0.19
0.18
0.17
noiseless (Suzuki-2)
E_static
0.28
0.26
0.24
0.22
0.20
0.18
8
10
12
14
time
[PLOT_CAPTION] Drive: energy vs time - top E_total, bottom E_static.
Exact vs noiseless (ADAPT-HVA+Suzuki-2) vs noisy (mode=ideal, 4096 shots).
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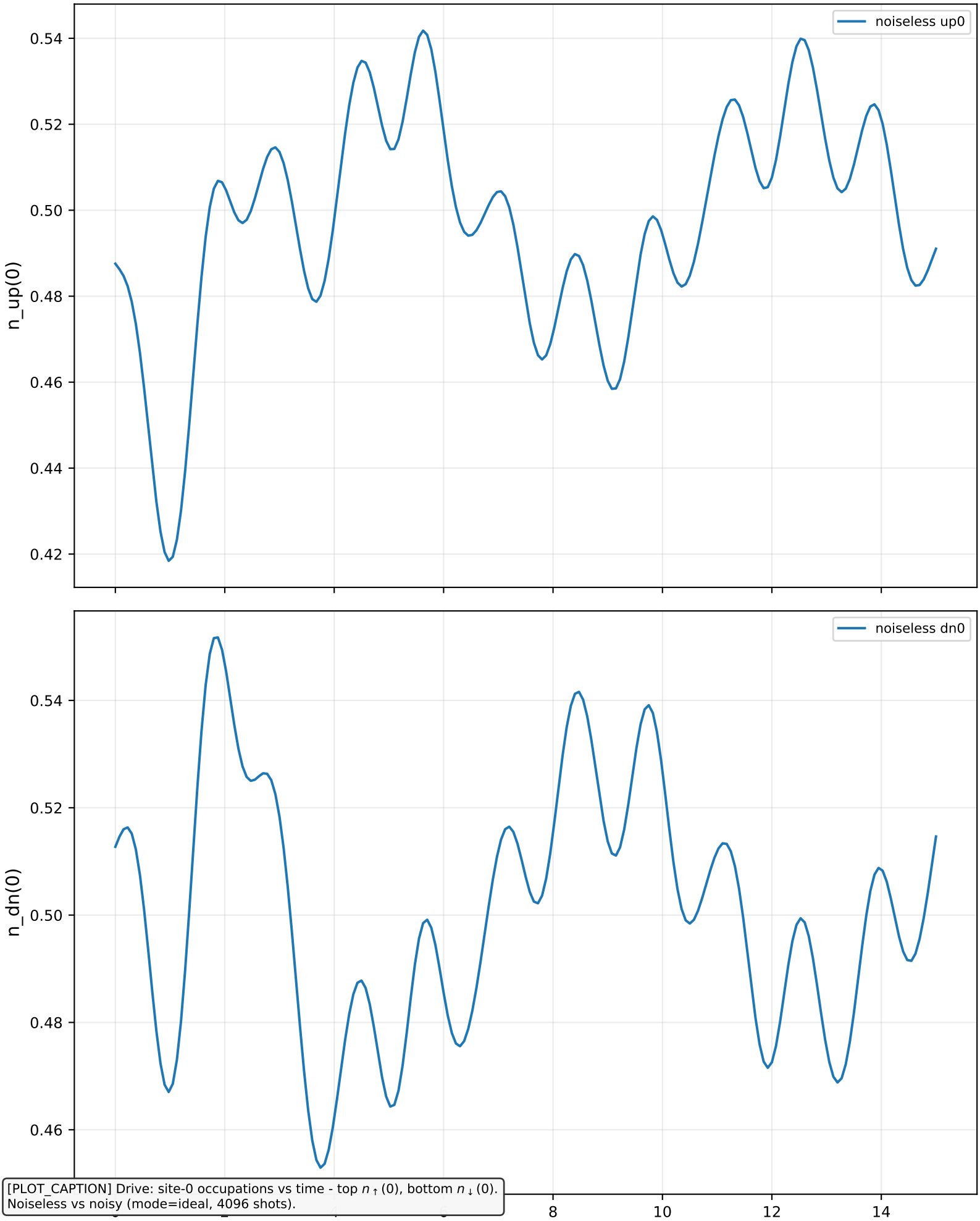
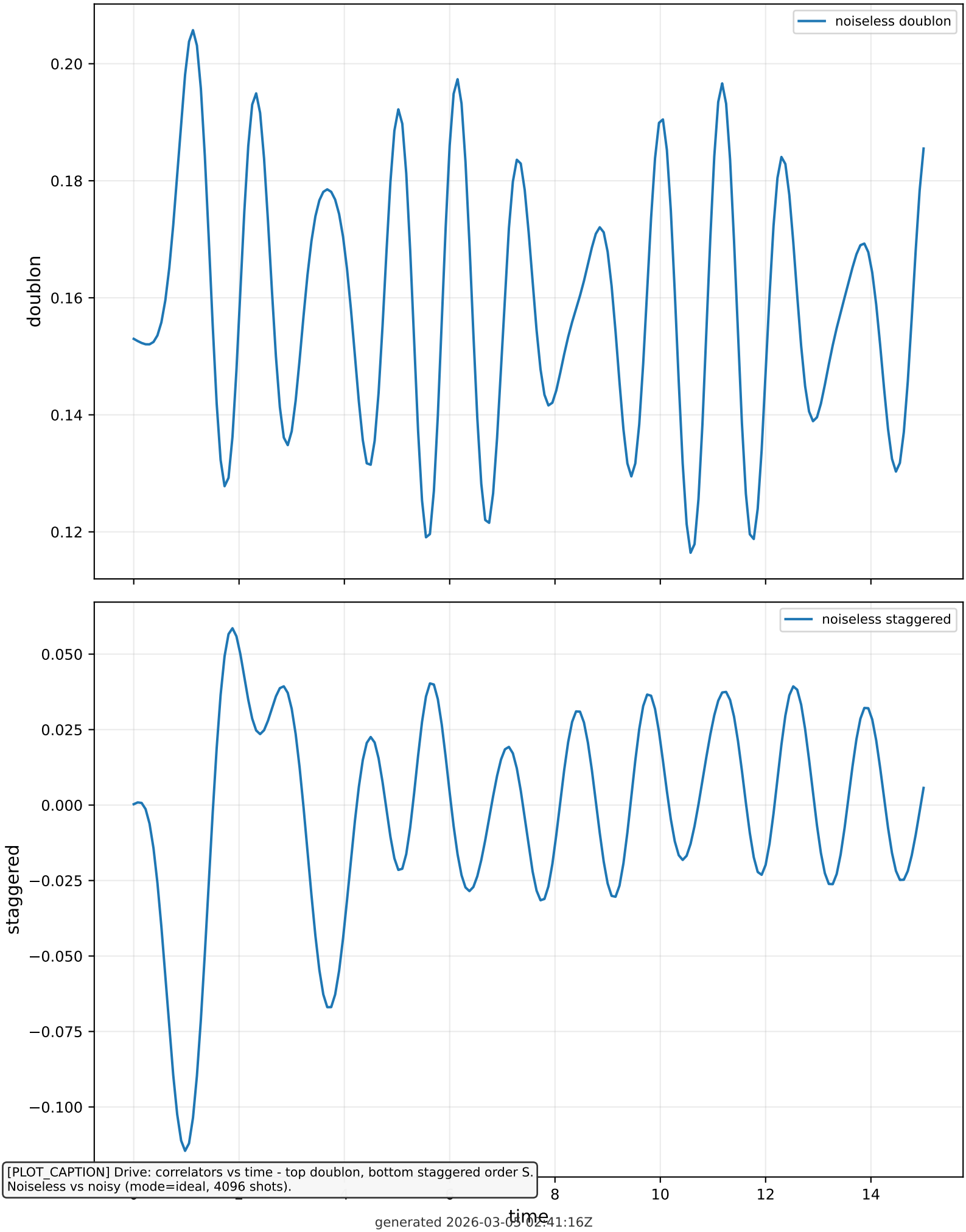
drive: site-0 occupation overlays

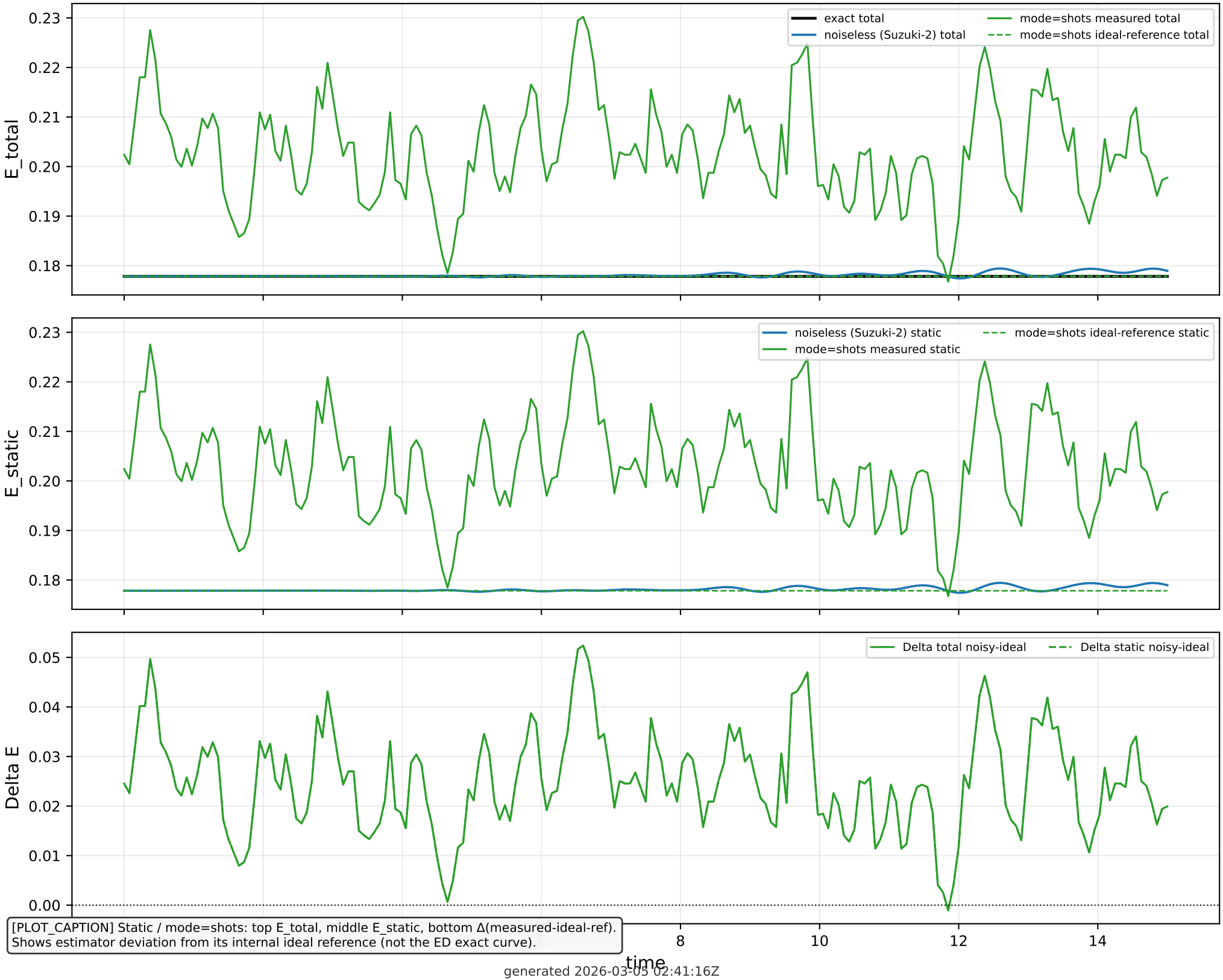


[PLOT_CAPTION] Drive: site-0 occupations vs time - top $n_\uparrow(0)$, bottom $n_\downarrow(0)$. Noiseless vs noisy (mode=ideal, 4096 shots).



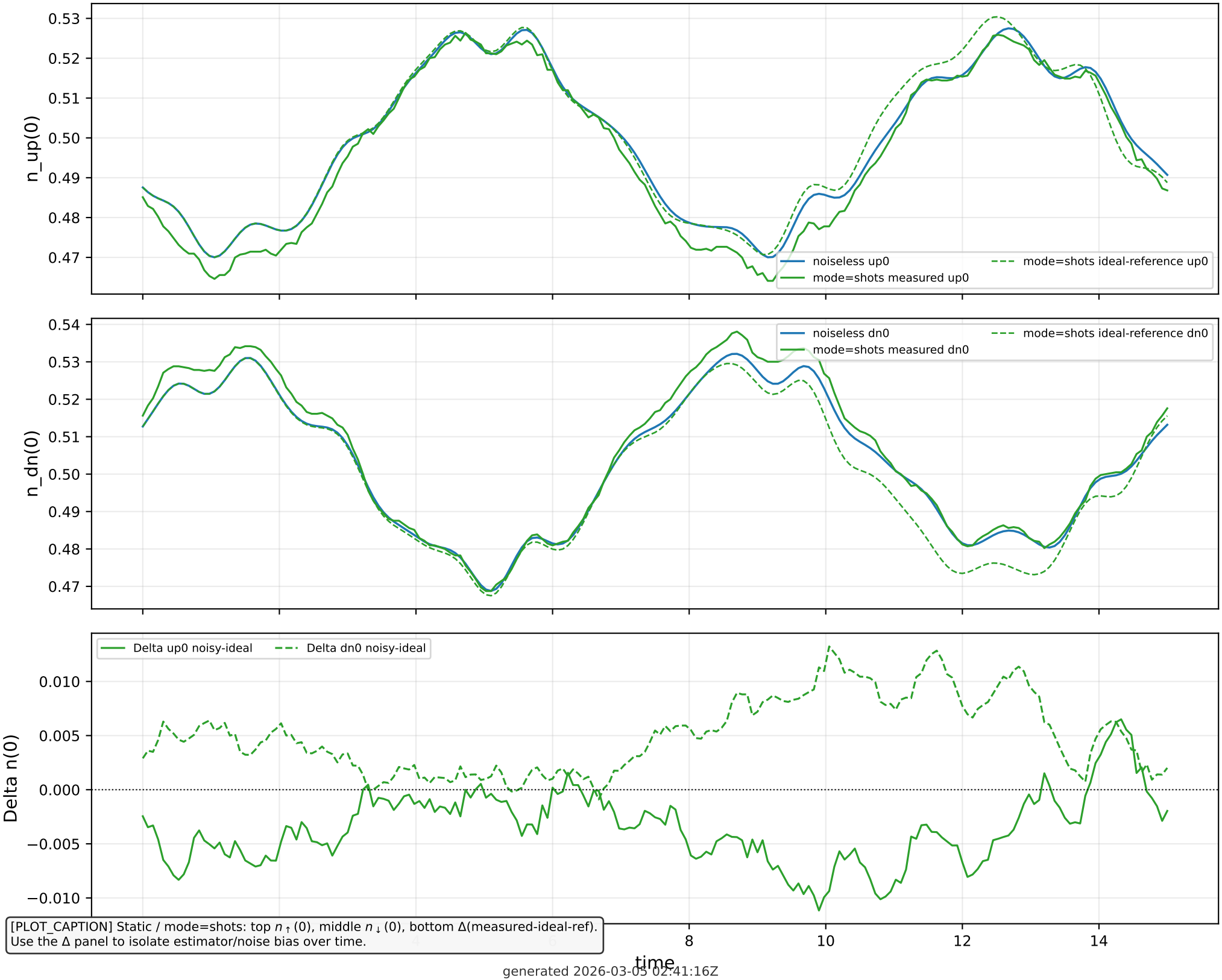
[PLOT_CAPTION] Drive: correlators vs time - top doublon, bottom staggered order S. Noiseless vs noisy (mode=ideal, 4096 shots).

static/shots: noisy energy audit



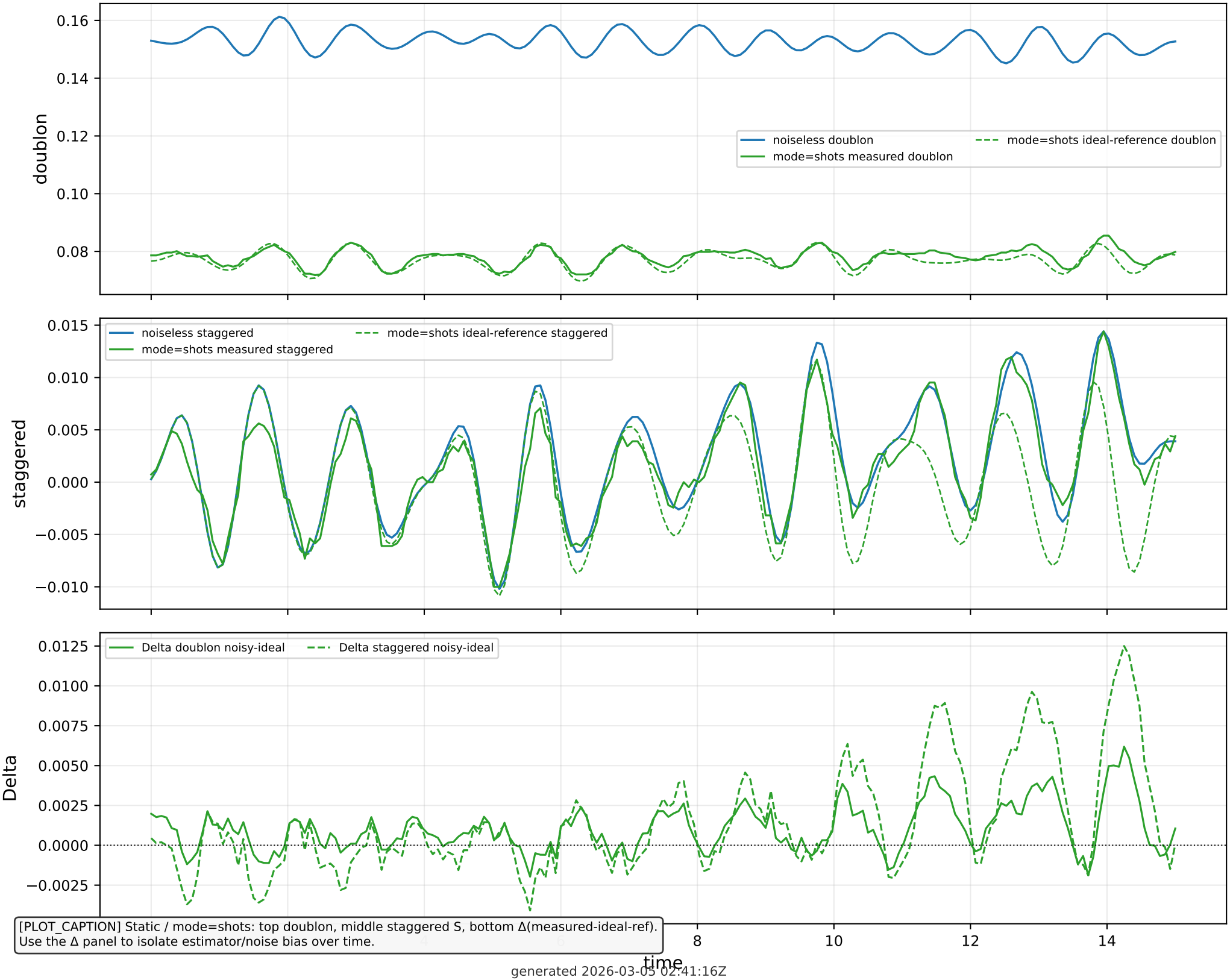
[PLOT_CAPTION] Static / mode=shots: top E_total, middle E_static, bottom Δ(measured-ideal-ref).
Shows estimator deviation from its internal ideal reference (not the ED exact curve).

static/shots: site-0 occupation audit



[PLOT_CAPTION] Static / mode=shots: top $n_\uparrow(0)$, middle $n_\downarrow(0)$, bottom Δ(measured-ideal-ref).
Use the Δ panel to isolate estimator/noise bias over time.

static/shots: doublon & staggered audit



[PLOT_CAPTION] Static / mode=shots: top doublon, middle staggered S, bottom Δ(measured-ideal-ref).
Use the Δ panel to isolate estimator/noise bias over time.

SECTION: NOISE MODE UNAVAILABLE (static/aer_noise)

Reason: timeout_after_21600s

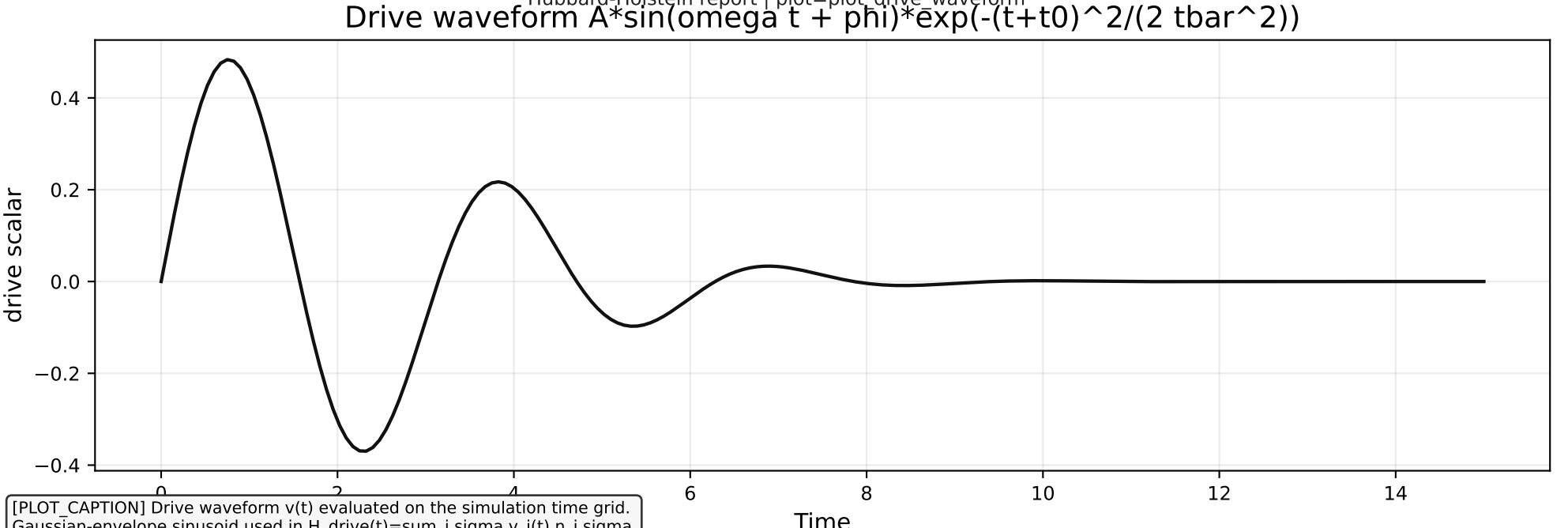
Diagnostics: {
  "success": false,
  "env_blocked": true,
  "reason": "timeout_after_21600s",
  "exitcode": -15
}

SECTION: NOISE MODE UNAVAILABLE (drive/shots)

Reason: worker_exception
Diagnostics: { "success": false, "env_blocked": true, "reason": "worker_exception", "error": "TypeError: _run_noisy_method_trajectory() missing 1 required keyword-only argument: 'mitigation_config'", "exitcode": 0 }

SECTION: NOISE MODE UNAVAILABLE (drive/aer_noise)

Reason: worker_exception
Diagnostics: { "success": false, "env_blocked": true, "reason": "worker_exception", "error": "TypeError: _run_noisy_method_trajectory() missing 1 required keyword-only argument: 'mitigation_config'", "exitcode": 0 }



[PLOT_CAPTION] Drive waveform v(t) evaluated on the simulation time grid.
Gaussian-envelope sinusoid used in H_drive(t)=sum_i,sigma v_i(t) n_i,sigma.

Noisy dynamics benchmark (Trotter vs CFQM under noise)

profile	method	mode	cx	term_exp	sq	depth	wall_s	oracle_s
static	suzuki2	shots	5120	4352	10496	4352	8224.796	8215.061

[PLOT_CAPTION] Noisy dynamics benchmark table: profile/method/mode rows.
Reports proxy costs (term-exp, cx, sq, depth) and runtime totals (wall/oracle).

```
Diagnostics and availability

Comparisons keys: ['noiseless_vs_exact', 'noise_vs_noiseless', 'noise_vs_noiseless_methods']
Noisy mode diagnostics: {   "static:suzuki2:shots": {     "success": true,     "env_blocked": false,     "reason":
    null,     "error": null   },   "static:suzuki2:aer_noise": {     "success": false,     "env_blocked": true,
    "reason": "timeout_after_21600s",     "error": null   },   "static:cfqm4:shots": {     "success": false,
    "env_blocked": true,     "reason": "worker_exception",     "error": "TypeError: _run_noisy_method_trajectory()
    missing 1 required keyword-only argument: 'mitigation_config'"   },   "static:cfqm4:aer_noise": {
    "success": false,     "env_blocked": true,     "reason": "worker_exception",     "error": "TypeError:
    _run_noisy_method_trajectory() missing 1 required keyword-only argument: 'mitigation_config'"   },
    "drive:suzuki2:shots": {     "success": false,     "env_blocked": true,     "reason": "worker_exception",
    "error": "TypeError: _run_noisy_method_trajectory() missing 1 required keyword-only argument:
    'mitigation_config'"   },   "drive:suzuki2:aer_noise": {     "success": false,     "env_blocked": true,
    "reason": "worker_exception",     "error": "TypeError: _run_noisy_method_trajectory() missing 1 required
    keyword-only argument: 'mitigation_config'"   },   "drive:cfqm4:shots": {     "success": false,
    "env_blocked": true,     "reason": "worker_exception",     "error": "TypeError: _run_noisy_method_trajectory()
    missing 1 required keyword-only argument: 'mitigation_config'"   },   "drive:cfqm4:aer_noise": {     "success":
    false,     "env_blocked": true,     "reason": "worker_exception",     "error": "TypeError:
    _run_noisy_method_trajectory() missing 1 required keyword-only argument: 'mitigation_config'"   } }
Noisy dynamics benchmark rows: 1

Metric formulas:
  energy_static(t) = <psi(t)|H_static|psi(t)>
  energy_total(t)  = <psi(t)|H_static + H_drive(t)|psi(t)>
  n_up_site0 = <n_{0,up}> , n_dn_site0 = <n_{0,dn}>
  doublon(site0) = <n_{0,up} n_{0,dn}>
  staggered = (1/L) * Sum_j (-1)^j <n_j>

equation_registry_count: 92
plot_contract_count: 43
```

SECTION: APPENDIX DEFINITIONS USED

Short equation definitions used in result pages.

SECTION: APPENDIX DEFINITIONS USED

Formula index for this report.

[eq_cfqm]

$$U_{\mathrm{CFQM}}(\Delta t) \approx \prod_m \exp[-i\, a_m \Delta t\, H(t + c_m \Delta t)]$$

Meaning: Commutator-free Magnus stage product (CFQM4/CFQM6).

Units: unitary

[eq_delta_e]

$$\Delta E_k = E_k - E_{\mathrm{exact,\,sector}},\ \ \delta_k = |\Delta E_k|$$

Meaning: Stage transition error metrics.

Units: energy

[eq_doublon]

$$D_i = \langle n_{i\uparrow}\, n_{i\downarrow} \rangle$$

Meaning: Onsite doublon expectation.

Units: dimensionless

[eq_drive_waveform]

$$v_i(t) = s_i A \sin[\omega(t + t_0) + \phi] \exp\left(-\frac{(t + t_0)^2}{2\tilde{t}^2}\right)$$

Meaning: Gaussian-envelope sinusoidal drive with spatial weights s_i.

Units: energy

[eq_energy_static]

$$E^b_{\mathrm{static}}(t) = \langle \psi_b(t) | H_0 | \psi_b(t) \rangle$$

Meaning: Static-Hamiltonian energy channel for branch b.

Units: energy

[eq_energy_total]

$$E^b_{\mathrm{total}}(t) = \langle \psi_b(t) | H_0 + H_{\mathrm{drive}}(t) | \psi_b(t) \rangle$$

Meaning: Instantaneous total energy under drive.

Units: energy

[eq_error_abs]

$$\varepsilon_X(t) = |X_{\mathrm{approx}}(t) - X_{\mathrm{exact}}(t)|$$

Meaning: Absolute trajectory error metric.

Units: observable units

[eq_h_drive_density]

$$H_{\mathrm{drive}}(t) = \sum_{i,\sigma} v_i(t)\, n_{i\sigma}$$

Meaning: Drive couples to onsite electron number operators.

Units: energy

SECTION: APPENDIX DEFINITIONS USED

Formula index for this report.

[eq_magnus2]

$$U_{\text{Magnus2}}(\Delta t) = \exp\left[-i\Delta t\, H\left(t + \frac{\Delta t}{2}\right)\right]$$

Meaning: Exponential midpoint (Magnus-2) reference approximation.

Units: unitary

[eq_noisy_delta]

$$\Delta_O^{\text{noise}-\text{ideal}}(t) = \mu_O^{\text{noise}}(t) - \mu_O^{\text{ideal}}(t)$$

Meaning: Noisy-vs-ideal observable delta.

Units: observable units

[eq_noisy_estimator]

$$\mu_O = \text{agg}(\{\langle O\rangle_r\}_{r=1}^{R}),\ \sigma_O = \text{std}(\{\langle O\rangle_r\}_{r=1}^{R})$$

Meaning: Noisy oracle aggregate and empirical spread.

Units: observable units

[eq_number_operator]

$$n_{i\sigma} = \frac{I - Z_{q(i,\sigma)}}{2}$$

Meaning: JW number-operator identity used for occupancy observables.

Units: dimensionless

[eq_proxy_cost]

$$\text{cx_proxy_term}(p) = 2\max(w(p) - 1, 0),\ \text{sq_proxy_term}(p) = 2\,xy(p) + 1$$

Meaning: Per-term hardware proxy definitions for dynamics benchmark totals.

Units: proxy-count

[eq_runtime_bench]

$$t_{\text{wall}} = t_{\text{build}} + t_{\text{oracle}} + \cdots$$

Meaning: Runtime decomposition for noisy dynamics evaluation.

Units: seconds

[eq_slope_switch]

$$\text{switch} \Leftrightarrow |\text{slope}(\delta_{k-w+1:k})| \le \varepsilon \text{ for } p \text{ consecutivecheckpoints}$$

Meaning: Windowed abs-slope plateau rule for stage transitions.

Units: energy/checkpoint

[eq_staggered]

$$S = \frac{1}{L}\sum_{i=0}^{L-1}(-1)^i\langle n_i\rangle$$

Meaning: Staggered charge-density order parameter.

Units: dimensionless

SECTION: APPENDIX DEFINITIONS USED

Formula index for this report.

[eq_subspace_fidelity]

$$F_{\mathrm{sub}}^{b}(t) = \langle \psi_b^{\mathrm{trot}}(t) | P_{\mathrm{GS}}(t) | \psi_b^{\mathrm{trot}}(t) \rangle$$

Meaning: Projected fidelity against sector-filtered exact GS manifold.

Units: probability

[eq_suzuki2]

$$U_{\mathrm{S2}}(\Delta t) \approx \prod_j e^{-i\frac{\Delta t}{2}H_j} \prod_j^{\mathrm{rev}} e^{-i\frac{\Delta t}{2}H_j}$$

Meaning: Second-order symmetric Suzuki product step.

Units: unitary

[eq_total_density]

$$n_i = \langle n_{i\uparrow} + n_{i\downarrow} \rangle$$

Meaning: Site-resolved total occupancy.

Units: particles

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_cfqm]

$$U_{\mathrm{CFQM}}(\Delta t) \approx \prod_m \exp[-i\, a_m \Delta t\, H(t + c_m \Delta t)]$$

Meaning: Commutator-free Magnus stage product (CFQM4/CFQM6).

Units: unitary

Source: dynamics_noiseless.profiles.*.methods.cfqm4, dynamics_noiseless.profiles.*.methods.cfqm6

[eq_delta_e]

$$\Delta E_k = E_k - E_{\mathrm{exact,\,sector}},\ \ \delta_k = |\Delta E_k|$$

Meaning: Stage transition error metrics.

Units: energy

Source: stage_pipeline.*.transition.delta_abs_trace

[eq_doublon]

$$D_i = \langle n_{i\uparrow} n_{i\downarrow} \rangle$$

Meaning: Onsite doublon expectation.

Units: dimensionless

Source: hardcoded_superset.profiles.*.trajectory.*.doublon_*

[eq_doublon_drive_adapt]

$$doublon_{drive}^{adapt}(t)$$

Meaning: doublon channel for drive profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_doublon_drive_final]

$$doublon_{drive}^{final}(t)$$

Meaning: doublon channel for drive profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_doublon_drive_warm]

$$doublon_{drive}^{warm}(t)$$

Meaning: doublon channel for drive profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_doublon_static_adapt]

$$doublon_{static}^{adapt}(t)$$

Meaning: doublon channel for static profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_doublon_static_final]

$$doublon_{static}^{final}(t)$$

Meaning: doublon channel for static profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_doublon_static_warm]

$$doublon_{static}^{warm}(t)$$

Meaning: doublon channel for static profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_drive_waveform]

$$v_i(t) = s_i A \sin[\omega(t + t_0) + \phi] \exp\left(-\frac{(t + t_0)^2}{2\bar{t}^2}\right)$$

Meaning: Gaussian-envelope sinusoidal drive with spatial weights s_i.

Units: energy

Source: settings.drive_profile.A, settings.drive_profile.omega, ... (+1 more)

[eq_energy_static]

$$E_{\text{static}}^{b}(t) = \langle \psi_b(t) | H_0 | \psi_b(t) \rangle$$

Meaning: Static-Hamiltonian energy channel for branch b.

Units: energy

Source: hardcoded_superset.profiles.*.trajectory.*.energy_static_*

[eq_energy_static_drive_adapt]

$$energy_static_{drive}^{adapt}(t)$$

Meaning: energy_static channel for drive profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_energy_static_drive_final]

$$energy_static_{drive}^{final}(t)$$

Meaning: energy_static channel for drive profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_energy_static_drive_warm]

$$energy_static_{drive}^{warm}(t)$$

Meaning: energy_static channel for drive profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_energy_static_static_adapt]

$$energy_static_{static}^{adapt}(t)$$

Meaning: energy_static channel for static profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_energy_static_static_final]

$$energy_static_{static}^{final}(t)$$

Meaning: energy_static channel for static profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_energy_static_static_warm]

$$energy_static_{static}^{warm}(t)$$

Meaning: energy_static channel for static profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_energy_total]

$$E_{\text{total}}^{b}(t) = \langle \psi_b(t) | H_0 + H_{\text{drive}}(t) | \psi_b(t) \rangle$$

Meaning: Instantaneous total energy under drive.

Units: energy

Source: hardcoded_superset.profiles.*.trajectory.*.energy_total_*

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_energy_total_drive_adapt]

$$energy_total_{drive}^{adapt}(t)$$

Meaning: energy_total channel for drive profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_energy_total_drive_final]

$$energy_total_{drive}^{final}(t)$$

Meaning: energy_total channel for drive profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_energy_total_drive_warm]

$$energy_total_{drive}^{warm}(t)$$

Meaning: energy_total channel for drive profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_energy_total_static_adapt]

$$energy_total_{static}^{adapt}(t)$$

Meaning: energy_total channel for static profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_energy_total_static_final]

$$energy_total_{static}^{final}(t)$$

Meaning: energy_total channel for static profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_energy_total_static_warm]

$$energy_total_{static}^{warm}(t)$$

Meaning: energy_total channel for static profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_error_abs]

$$\varepsilon_X(t) = |X_{\text{approx}}(t) - X_{\text{exact}}(t)|$$

Meaning: Absolute trajectory error metric.

Units: observable units

Source: comparisons.noiseless_vs_exact, comparisons.noise_vs_noiseless

[eq_h_drive_density]

$$H_{\text{drive}}(t) = \sum_{i,\sigma} v_i(t)\, n_{i\sigma}$$

Meaning: Drive couples to onsite electron number operators.

Units: energy

Source: settings.drive_profile, diagnostics.metric_definitions.energy_total

[eq_h_total]

$$H(t) = H_0 + H_{\text{drive}}(t)$$

Meaning: Total Hamiltonian splits into static + drive terms.

Units: energy

Source: settings.t, settings.u, ... (+2 more)

[eq_magnus2]

$$U_{\text{Magnus2}}(\Delta t) = \exp\left[-i\Delta t\, H\left(t + \frac{\Delta t}{2}\right)\right]$$

Meaning: Exponential midpoint (Magnus-2) reference approximation.

Units: unitary

Source: dynamics_noiseless.profiles.*.methods.magnus2

[eq_n_dn_site0_drive_adapt]

$$n_d n_s ite0_{drive}^{adapt}(t)$$

Meaning: n_dn_site0 channel for drive profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_n_dn_site0_drive_final]

$$n_d n_s ite0_{drive}^{final}(t)$$

Meaning: n_dn_site0 channel for drive profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_n_dn_site0_drive_warm]

$$n_d n_s ite0_{drive}^{warm}(t)$$

Meaning: n_dn_site0 channel for drive profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_n_dn_site0_static_adapt]

$$n_d n_s ite0_{static}^{adapt}(t)$$

Meaning: n_dn_site0 channel for static profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_n_dn_site0_static_final]

$$n_d n_s ite0_{static}^{final}(t)$$

Meaning: n_dn_site0 channel for static profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_n_dn_site0_static_warm]

$$n_d n_s ite0_{static}^{warm}(t)$$

Meaning: n_dn_site0 channel for static profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_n_up_site0_drive_adapt]

$$n_u p_s ite0_{drive}^{adapt}(t)$$

Meaning: n_up_site0 channel for drive profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_n_up_site0_drive_final]

$$n_u p_s ite0_{drive}^{final}(t)$$

Meaning: n_up_site0 channel for drive profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_n_up_site0_drive_warm]

$$n_u p_s ite0_{drive}^{warm}(t)$$

Meaning: n_up_site0 channel for drive profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_n_up_site0_static_adapt]

$$n_u p_s ite0_{static}^{adapt}(t)$$

Meaning: n_up_site0 channel for static profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_n_up_site0_static_final]

$$n_u p_s ite0_{static}^{final}(t)$$

Meaning: n_up_site0 channel for static profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_n_up_site0_static_warm]

$$n_u p_s ite0_{static}^{warm}(t)$$

Meaning: n_up_site0 channel for static profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_noisy_delta]

$$\Delta_O^{\mathrm{noise-ideal}}(t) = \mu_O^{\mathrm{noise}}(t) - \mu_O^{\mathrm{ideal}}(t)$$

Meaning: Noisy-vs-ideal observable delta.

Units: observable units

Source: dynamics_noisy.profiles.*.modes.*.trajectory.*.*_delta_noisy_minus_ideal

[eq_noisy_doublon_drive_aer_noise]

$$\mu_{doublon}^{aer_noise,\,drive}(t)$$

Meaning: Noisy estimate for doublon in drive/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.aer_noise.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_noisy_doublon_drive_ideal]

$\mu_{doublon}^{ideal,\,drive}(t)$

Meaning: Noisy estimate for doublon in drive/ideal.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.ideal.trajectory

[eq_noisy_doublon_drive_shots]

$\mu_{doublon}^{shots,\,drive}(t)$

Meaning: Noisy estimate for doublon in drive/shots.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.shots.trajectory

[eq_noisy_doublon_static_aer_noise]

$\mu_{doublon}^{aer_noise,\,static}(t)$

Meaning: Noisy estimate for doublon in static/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.aer_noise.trajectory

[eq_noisy_doublon_static_ideal]

$\mu_{doublon}^{ideal,\,static}(t)$

Meaning: Noisy estimate for doublon in static/ideal.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.ideal.trajectory

[eq_noisy_doublon_static_shots]

$\mu_{doublon}^{shots,\,static}(t)$

Meaning: Noisy estimate for doublon in static/shots.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.shots.trajectory

[eq_noisy_energy_static_drive_aer_noise]

$\mu_{energy_static}^{aer_noise,\,drive}(t)$

Meaning: Noisy estimate for energy_static in drive/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.aer_noise.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_noisy_energy_static_drive_ideal]

$\mu_{energy_static}^{ideal,\,drive}(t)$

Meaning: Noisy estimate for energy_static in drive/ideal.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.ideal.trajectory

[eq_noisy_energy_static_drive_shots]

$\mu_{energy_static}^{shots,\,drive}(t)$

Meaning: Noisy estimate for energy_static in drive/shots.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.shots.trajectory

[eq_noisy_energy_static_static_aer_noise]

$\mu_{energy_static}^{aer_noise,\,static}(t)$

Meaning: Noisy estimate for energy_static in static/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.aer_noise.trajectory

[eq_noisy_energy_static_static_ideal]

$\mu_{energy_static}^{ideal,\,static}(t)$

Meaning: Noisy estimate for energy_static in static/ideal.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.ideal.trajectory

[eq_noisy_energy_static_static_shots]

$\mu_{energy_static}^{shots,\,static}(t)$

Meaning: Noisy estimate for energy_static in static/shots.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.shots.trajectory

[eq_noisy_energy_total_drive_aer_noise]

$\mu_{energy_total}^{aer_noise,\,drive}(t)$

Meaning: Noisy estimate for energy_total in drive/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.aer_noise.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_noisy_energy_total_drive_ideal]

$$\mu^{ideal,\,drive}_{energy_total}(t)$$

Meaning: Noisy estimate for energy_total in drive/ideal.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.ideal.trajectory

[eq_noisy_energy_total_drive_shots]

$$\mu^{shots,\,drive}_{energy_total}(t)$$

Meaning: Noisy estimate for energy_total in drive/shots.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.shots.trajectory

[eq_noisy_energy_total_static_aer_noise]

$$\mu^{aer_noise,\,static}_{energy_total}(t)$$

Meaning: Noisy estimate for energy_total in static/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.aer_noise.trajectory

[eq_noisy_energy_total_static_ideal]

$$\mu^{ideal,\,static}_{energy_total}(t)$$

Meaning: Noisy estimate for energy_total in static/ideal.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.ideal.trajectory

[eq_noisy_energy_total_static_shots]

$$\mu^{shots,\,static}_{energy_total}(t)$$

Meaning: Noisy estimate for energy_total in static/shots.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.shots.trajectory

[eq_noisy_estimator]

$$\mu_O = \mathrm{agg}(\{\langle O\rangle_r\}_{r=1}^{R}),\ \sigma_O = \mathrm{std}(\{\langle O\rangle_r\}_{r=1}^{R})$$

Meaning: Noisy oracle aggregate and empirical spread.

Units: observable units

Source: settings.oracle_repeats, settings.oracle_aggregate

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_noisy_n_dn_site0_drive_aer_noise]

$\mu_{n_d n_s ite0}^{aer_n oise,\, drive}(t)$

Meaning: Noisy estimate for n_dn_site0 in drive/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.aer_noise.trajectory

[eq_noisy_n_dn_site0_drive_ideal]

$\mu_{n_d n_s ite0}^{ideal,\, drive}(t)$

Meaning: Noisy estimate for n_dn_site0 in drive/ideal.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.ideal.trajectory

[eq_noisy_n_dn_site0_drive_shots]

$\mu_{n_d n_s ite0}^{shots,\, drive}(t)$

Meaning: Noisy estimate for n_dn_site0 in drive/shots.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.shots.trajectory

[eq_noisy_n_dn_site0_static_aer_noise]

$\mu_{n_d n_s ite0}^{aer_n oise,\, static}(t)$

Meaning: Noisy estimate for n_dn_site0 in static/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.aer_noise.trajectory

[eq_noisy_n_dn_site0_static_ideal]

$\mu_{n_d n_s ite0}^{ideal,\, static}(t)$

Meaning: Noisy estimate for n_dn_site0 in static/ideal.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.ideal.trajectory

[eq_noisy_n_dn_site0_static_shots]

$\mu_{n_d n_s ite0}^{shots,\, static}(t)$

Meaning: Noisy estimate for n_dn_site0 in static/shots.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.shots.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_noisy_n_up_site0_drive_aer_noise]

$\mu_{n_u p_s ite0}^{aer_n oise,\, drive}(t)$

Meaning: Noisy estimate for n_up_site0 in drive/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.aer_noise.trajectory

[eq_noisy_n_up_site0_drive_ideal]

$\mu_{n_u p_s ite0}^{ideal,\, drive}(t)$

Meaning: Noisy estimate for n_up_site0 in drive/ideal.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.ideal.trajectory

[eq_noisy_n_up_site0_drive_shots]

$\mu_{n_u p_s ite0}^{shots,\, drive}(t)$

Meaning: Noisy estimate for n_up_site0 in drive/shots.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.shots.trajectory

[eq_noisy_n_up_site0_static_aer_noise]

$\mu_{n_u p_s ite0}^{aer_n oise,\, static}(t)$

Meaning: Noisy estimate for n_up_site0 in static/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.aer_noise.trajectory

[eq_noisy_n_up_site0_static_ideal]

$\mu_{n_u p_s ite0}^{ideal,\, static}(t)$

Meaning: Noisy estimate for n_up_site0 in static/ideal.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.ideal.trajectory

[eq_noisy_n_up_site0_static_shots]

$\mu_{n_u p_s ite0}^{shots,\, static}(t)$

Meaning: Noisy estimate for n_up_site0 in static/shots.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.shots.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_noisy_staggered_drive_aer_noise]

$\mu_{staggered}^{aer_noise,\,drive}(t)$

Meaning: Noisy estimate for staggered in drive/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.aer_noise.trajectory

[eq_noisy_staggered_drive_ideal]

$\mu_{staggered}^{ideal,\,drive}(t)$

Meaning: Noisy estimate for staggered in drive/ideal.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.ideal.trajectory

[eq_noisy_staggered_drive_shots]

$\mu_{staggered}^{shots,\,drive}(t)$

Meaning: Noisy estimate for staggered in drive/shots.

Units: observable units

Source: dynamics_noisy.profiles.drive.modes.shots.trajectory

[eq_noisy_staggered_static_aer_noise]

$\mu_{staggered}^{aer_noise,\,static}(t)$

Meaning: Noisy estimate for staggered in static/aer_noise.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.aer_noise.trajectory

[eq_noisy_staggered_static_ideal]

$\mu_{staggered}^{ideal,\,static}(t)$

Meaning: Noisy estimate for staggered in static/ideal.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.ideal.trajectory

[eq_noisy_staggered_static_shots]

$\mu_{staggered}^{shots,\,static}(t)$

Meaning: Noisy estimate for staggered in static/shots.

Units: observable units

Source: dynamics_noisy.profiles.static.modes.shots.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_number_operator]

$$n_{i\sigma} = \frac{I - Z_{q(i,\sigma)}}{2}$$

Meaning: JW number-operator identity used for occupancy observables.

Units: dimensionless

Source: diagnostics.metric_definitions

[eq_proxy_cost]

$$\text{cx_proxy_term}(p) = 2\max(w(p) - 1, 0),\ \text{sq_proxy_term}(p) = 2\,xy(p) + 1$$

Meaning: Per-term hardware proxy definitions for dynamics benchmark totals.

Units: proxy-count

Source: dynamics_noisy.profiles.*.methods.*.modes.*.benchmark_cost

[eq_runtime_bench]

$$t_{\text{wall}} = t_{\text{build}} + t_{\text{oracle}} + \cdots$$

Meaning: Runtime decomposition for noisy dynamics evaluation.

Units: seconds

Source: dynamics_noisy.profiles.*.methods.*.modes.*.benchmark_runtime

[eq_slope_switch]

$$\text{switch} \Leftrightarrow |\text{slope}(\delta_{k-w+1:k})| \le \varepsilon \text{ for } p \text{ consecutivecheckpoints}$$

Meaning: Windowed abs-slope plateau rule for stage transitions.

Units: energy/checkpoint

Source: settings.transition_policy

[eq_staggered]

$$S = \frac{1}{L}\sum_{i=0}^{L-1}(-1)^i\langle n_i\rangle$$

Meaning: Staggered charge-density order parameter.

Units: dimensionless

Source: hardcoded_superset.profiles.*.trajectory.*.staggered_*

[eq_staggered_drive_adapt]

$$staggered_{drive}^{adapt}(t)$$

Meaning: staggered channel for drive profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_staggered_drive_final]

$$staggered_{drive}^{final}(t)$$

Meaning: staggered channel for drive profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_staggered_drive_warm]

$$staggered_{drive}^{warm}(t)$$

Meaning: staggered channel for drive profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.drive.trajectory

[eq_staggered_static_adapt]

$$staggered_{static}^{adapt}(t)$$

Meaning: staggered channel for static profile, adapt branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_staggered_static_final]

$$staggered_{static}^{final}(t)$$

Meaning: staggered channel for static profile, final branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_staggered_static_warm]

$$staggered_{static}^{warm}(t)$$

Meaning: staggered channel for static profile, warm branch.

Units: observable units

Source: hardcoded_superset.profiles.static.trajectory

[eq_subspace_fidelity]

$$F_{\mathrm{sub}}^{b}(t) = \langle \psi_b^{\mathrm{trot}}(t) | P_{\mathrm{GS}}(t) | \psi_b^{\mathrm{trot}}(t) \rangle$$

Meaning: Projected fidelity against sector-filtered exact GS manifold.

Units: probability

Source: hardcoded_superset.profiles.*.trajectory.*.fidelity*

SECTION: APPENDIX FORMULA ATLAS

Formula index for this report.

[eq_suzuki2]

$$U_{S2}(\Delta t) \approx \prod_j e^{-i\frac{\Delta t}{2}H_j} \prod_j^{\text{rev}} e^{-i\frac{\Delta t}{2}H_j}$$

Meaning: Second-order symmetric Suzuki product step.

Units: unitary

Source: dynamics_noiseless.profiles.*.methods.suzuki2

[eq_total_density]

$$n_i = \langle n_{i\uparrow} + n_{i\downarrow} \rangle$$

Meaning: Site-resolved total occupancy.

Units: particles

Source: hardcoded_superset.profiles.*.trajectory.*.n_site_*

SECTION: APPENDIX PLOT CONTRACTS

Full machine-readable plot contract map.

```
SECTION: APPENDIX PLOT CONTRACTS

[plot_drive_aer_noise_doublon_staggered]
  x: time grid
  y: ['doublon_noisy', 'doublon_ideal', 'staggered_noisy', 'staggered_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.aer_noise.trajectory']
  notes: Noisy doublon/staggered channels and deltas.

[plot_drive_aer_noise_energy]
  x: time grid
  y: ['energy_static_noisy', 'energy_static_ideal', 'energy_total_noisy', 'energy_total_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.aer_noise.trajectory']
  notes: Noisy energy channels and noisy-ideal deltas.

[plot_drive_aer_noise_occupancy]
  x: time grid
  y: ['n_up_site0_noisy', 'n_up_site0_ideal', 'n_dn_site0_noisy', 'n_dn_site0_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.aer_noise.trajectory']
  notes: Noisy site-0 spin channels and deltas.

[plot_drive_branch_overlay_doublon_staggered]
  x: time grid
  y: ['doublon_trotter', 'doublon_trotter_paop', 'doublon_trotter_hva', 'staggered_trotter', 'staggered_trotter_paop',
    'staggered_trotter_hva']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Three-branch mapped doublon/staggered overlay.

[plot_drive_branch_overlay_energy]
  x: time grid
  y: ['energy_total_trotter', 'energy_total_trotter_paop', 'energy_total_trotter_hva', 'energy_total_exact']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Three-branch mapped overlay (warm/adapt/final).

[plot_drive_density_dn_heatmaps]
  x: time grid
  y: ['n_dn_site_exact', 'n_dn_site_exact_hva', 'n_dn_site_trotter_hva']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Spin-down density family.
```

```
[plot_drive_density_total_heatmaps]
  x: time grid
  y: ['n_site_exact', 'n_site_exact_hva', 'n_site_trotter_hva']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Heatmaps: exact GS, exact final branch, trotter final branch.

[plot_drive_density_up_heatmaps]
  x: time grid
  y: ['n_up_site_exact', 'n_up_site_exact_hva', 'n_up_site_trotter_hva']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Spin-up density family.
```

SECTION: APPENDIX PLOT CONTRACTS (cont. 8)

```
[plot_drive_energy_fidelity_final]
  x: time grid
  y: ['fidelity_hva_trotter', 'energy_total_exact', 'energy_total_trotter_hva', 'doublon_exact_hva', 'doublon_trotter_hva']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Final-branch hardcoded family summary.

[plot_drive_ideal_doublon_staggered]
  x: time grid
  y: ['doublon_noisy', 'doublon_ideal', 'staggered_noisy', 'staggered_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.ideal.trajectory']
  notes: Noisy doublon/staggered channels and deltas.

[plot_drive_ideal_energy]
  x: time grid
  y: ['energy_static_noisy', 'energy_static_ideal', 'energy_total_noisy', 'energy_total_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.ideal.trajectory']
  notes: Noisy energy channels and noisy-ideal deltas.

[plot_drive_ideal_occupancy]
  x: time grid
  y: ['n_up_site0_noisy', 'n_up_site0_ideal', 'n_dn_site0_noisy', 'n_dn_site0_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.ideal.trajectory']
  notes: Noisy site-0 spin channels and deltas.

[plot_drive_magnus_cfqm_error]
  x: time grid
  y: ['|E_method-E_exact|', '|F_method-1|']
  source: ['dynamics_noiseless.profiles.drive.methods.*.trajectory']
  notes: Noiseless error channels vs exact.

[plot_drive_magnus_cfqm_overlay]
  x: time grid
  y: ['suzuki2.energy_total_trotter', 'magnus2.energy_total_trotter', 'cfqm4.energy_total_trotter', 'cfqm6.energy_total_trotter',
    'exact.energy_total_exact']
  source: ['dynamics_noiseless.profiles.drive.methods.*.trajectory']
  notes: Noiseless method comparison.
```

```
[plot_drive_overlay_doublon_staggered]
  x: time grid
  y: ['doublon', 'staggered overlays']
  source: ['dynamics_noisy.profiles.drive.modes.*.trajectory', 'dynamics_noiseless.profiles.drive.methods.suzuki2.trajectory']
  notes: Overlay page for doublon and staggered order.

[plot_drive_overlay_energy]
  x: time grid
  y: ['noiseless+suzuki2+exact+noisy energy overlays']
  source: ['dynamics_noisy.profiles.drive.modes.*.trajectory', 'dynamics_noiseless.profiles.drive.methods.suzuki2.trajectory']
  notes: Overlay page for exact/noiseless/noisy energies.
```

SECTION: APPENDIX PLOT CONTRACTS (cont. 16)

```
[plot_drive_overlay_occupancy]
  x: time grid
  y: ['n_up(0)', 'n_dn(0) overlays']
  source: ['dynamics_noisy.profiles.drive.modes.*.trajectory', 'dynamics_noiseless.profiles.drive.methods.suzuki2.trajectory']
  notes: Overlay page for site-0 occupancies.

[plot_drive_scalar_error_heatmap]
  x: time grid
  y: ['|E_*| rows', '|D_*| rows', '|S_*| rows']
  source: ['hardcoded_superset.profiles.drive.trajectory']
  notes: Absolute scalar error channels heatmap.

[plot_drive_shots_doublon_staggered]
  x: time grid
  y: ['doublon_noisy', 'doublon_ideal', 'staggered_noisy', 'staggered_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.shots.trajectory']
  notes: Noisy doublon/staggered channels and deltas.

[plot_drive_shots_energy]
  x: time grid
  y: ['energy_static_noisy', 'energy_static_ideal', 'energy_total_noisy', 'energy_total_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.shots.trajectory']
  notes: Noisy energy channels and noisy-ideal deltas.

[plot_drive_shots_occupancy]
  x: time grid
  y: ['n_up_site0_noisy', 'n_up_site0_ideal', 'n_dn_site0_noisy', 'n_dn_site0_ideal']
  source: ['dynamics_noisy.profiles.drive.modes.shots.trajectory']
  notes: Noisy site-0 spin channels and deltas.

[plot_stage_transition]
  x: checkpoint index
  y: ['delta_abs_trace', 'slope_trace']
  source: ['stage_pipeline.warm_start.transition', 'stage_pipeline.adapt_pool_b.transition']
  notes: Windowed abs-slope switching traces.

[plot_static_aer_noise_doublon_staggered]
```

```
  x: time grid
  y: ['doublon_noisy', 'doublon_ideal', 'staggered_noisy', 'staggered_ideal']
  source: ['dynamics_noisy.profiles.static.modes.aer_noise.trajectory']
  notes: Noisy doublon/staggered channels and deltas.

[plot_static_aer_noise_energy]
  x: time grid
  y: ['energy_static_noisy', 'energy_static_ideal', 'energy_total_noisy', 'energy_total_ideal']
  source: ['dynamics_noisy.profiles.static.modes.aer_noise.trajectory']
  notes: Noisy energy channels and noisy-ideal deltas.
```

SECTION: APPENDIX PLOT CONTRACTS (cont. 24)

```
[plot_static_aer_noise_occupancy]
  x: time grid
  y: ['n_up_site0_noisy', 'n_up_site0_ideal', 'n_dn_site0_noisy', 'n_dn_site0_ideal']
  source: ['dynamics_noisy.profiles.static.modes.aer_noise.trajectory']
  notes: Noisy site-0 spin channels and deltas.

[plot_static_branch_overlay_doublon_staggered]
  x: time grid
  y: ['doublon_trotter', 'doublon_trotter_paop', 'doublon_trotter_hva', 'staggered_trotter', 'staggered_trotter_paop',
    'staggered_trotter_hva']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Three-branch mapped doublon/staggered overlay.

[plot_static_branch_overlay_energy]
  x: time grid
  y: ['energy_total_trotter', 'energy_total_trotter_paop', 'energy_total_trotter_hva', 'energy_total_exact']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Three-branch mapped overlay (warm/adapt/final).

[plot_static_density_dn_heatmaps]
  x: time grid
  y: ['n_dn_site_exact', 'n_dn_site_exact_hva', 'n_dn_site_trotter_hva']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Spin-down density family.

[plot_static_density_total_heatmaps]
  x: time grid
  y: ['n_site_exact', 'n_site_exact_hva', 'n_site_trotter_hva']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Heatmaps: exact GS, exact final branch, trotter final branch.

[plot_static_density_up_heatmaps]
  x: time grid
  y: ['n_up_site_exact', 'n_up_site_exact_hva', 'n_up_site_trotter_hva']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Spin-up density family.
```

```
[plot_static_energy_fidelity_final]
  x: time grid
  y: ['fidelity_hva_trotter', 'energy_total_exact', 'energy_total_trotter_hva', 'doublon_exact_hva', 'doublon_trotter_hva']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Final-branch hardcoded family summary.

[plot_static_ideal_doublon_staggered]
  x: time grid
  y: ['doublon_noisy', 'doublon_ideal', 'staggered_noisy', 'staggered_ideal']
  source: ['dynamics_noisy.profiles.static.modes.ideal.trajectory']
  notes: Noisy doublon/staggered channels and deltas.
```

SECTION: APPENDIX PLOT CONTRACTS (cont. 32)

```
[plot_static_ideal_energy]
  x: time grid
  y: ['energy_static_noisy', 'energy_static_ideal', 'energy_total_noisy', 'energy_total_ideal']
  source: ['dynamics_noisy.profiles.static.modes.ideal.trajectory']
  notes: Noisy energy channels and noisy-ideal deltas.

[plot_static_ideal_occupancy]
  x: time grid
  y: ['n_up_site0_noisy', 'n_up_site0_ideal', 'n_dn_site0_noisy', 'n_dn_site0_ideal']
  source: ['dynamics_noisy.profiles.static.modes.ideal.trajectory']
  notes: Noisy site-0 spin channels and deltas.

[plot_static_magnus_cfqm_error]
  x: time grid
  y: ['|E_method-E_exact|', '|F_method-1|']
  source: ['dynamics_noiseless.profiles.static.methods.*.trajectory']
  notes: Noiseless error channels vs exact.

[plot_static_magnus_cfqm_overlay]
  x: time grid
  y: ['suzuki2.energy_total_trotter', 'magnus2.energy_total_trotter', 'cfqm4.energy_total_trotter', 'cfqm6.energy_total_trotter',
    'exact.energy_total_exact']
  source: ['dynamics_noiseless.profiles.static.methods.*.trajectory']
  notes: Noiseless method comparison.

[plot_static_overlay_doublon_staggered]
  x: time grid
  y: ['doublon', 'staggered overlays']
  source: ['dynamics_noisy.profiles.static.modes.*.trajectory', 'dynamics_noiseless.profiles.static.methods.suzuki2.trajectory']
  notes: Overlay page for doublon and staggered order.

[plot_static_overlay_energy]
  x: time grid
  y: ['noiseless+suzuki2+exact+noisy energy overlays']
  source: ['dynamics_noisy.profiles.static.modes.*.trajectory', 'dynamics_noiseless.profiles.static.methods.suzuki2.trajectory']
  notes: Overlay page for exact/noiseless/noisy energies.
```

```
[plot_static_overlay_occupancy]
  x: time grid
  y: ['n_up(0)', 'n_dn(0) overlays']
  source: ['dynamics_noisy.profiles.static.modes.*.trajectory', 'dynamics_noiseless.profiles.static.methods.suzuki2.trajectory']
  notes: Overlay page for site-0 occupancies.

[plot_static_scalar_error_heatmap]
  x: time grid
  y: ['|E_*| rows', '|D_*| rows', '|S_*| rows']
  source: ['hardcoded_superset.profiles.static.trajectory']
  notes: Absolute scalar error channels heatmap.
```

SECTION: APPENDIX PLOT CONTRACTS (cont. 40)

```
[plot_static_shots_doublon_staggered]
  x: time grid
  y: ['doublon_noisy', 'doublon_ideal', 'staggered_noisy', 'staggered_ideal']
  source: ['dynamics_noisy.profiles.static.modes.shots.trajectory']
  notes: Noisy doublon/staggered channels and deltas.

[plot_static_shots_energy]
  x: time grid
  y: ['energy_static_noisy', 'energy_static_ideal', 'energy_total_noisy', 'energy_total_ideal']
  source: ['dynamics_noisy.profiles.static.modes.shots.trajectory']
  notes: Noisy energy channels and noisy-ideal deltas.

[plot_static_shots_occupancy]
  x: time grid
  y: ['n_up_site0_noisy', 'n_up_site0_ideal', 'n_dn_site0_noisy', 'n_dn_site0_ideal']
  source: ['dynamics_noisy.profiles.static.modes.shots.trajectory']
  notes: Noisy site-0 spin channels and deltas.
```

Executed Command

Reference: pipelines/run_guide.md
Script: pipelines/exact_bench/hh_noise_robustness_seq_report.py

```
/opt/anaconda3/bin/python /Users/jakestrobel/Documents/Holstein_implementation/Holstein_test/pipelines/exact_b
    ench/hh_noise_robustness_seq_report.py --L 2 --t 1.0 --u 4.0 --dv 0.0 --omega0 1.0 --g-ep 0.5 --n-ph-max 1
    --boundary open --ordering blocked --warm-reps 2 --warm-restarts 4 --warm-maxiter 2400 --warm-method
    COBYLA --warm-seed 7 --adapt-max-depth 80 --adapt-maxiter 3000 --adapt-eps-grad 5e-7 --adapt-eps-energy
    1e-9 --adapt-seed 7 --adapt-allow-repeats --final-reps 3 --final-restarts 4 --final-maxiter 2400 --final-
    method COBYLA --final-seed 7 --t-final 15.0 --num-times 201 --trotter-steps 128 --exact-steps-multiplier 2
    --include-drive-profile --drive-A 0.5 --drive-omega 2.0 --drive-tbar 3.0 --drive-phi 0.0 --drive-pattern
    staggered --drive-time-sampling midpoint --noise-modes shots,aer_noise --noisy-methods suzuki2,cfqm4
    --shots 4096 --oracle-repeats 6 --oracle-aggregate mean --noise-seed 7 --use-fake-backend --backend-name
    FakeManilaV2 --no-allow-aer-fallback --omp-shm-workaround --noisy-mode-timeout-s 21600 --output-json /User
    s/jakestrobel/Documents/Holstein_implementation/Holstein_test/artifacts/json/hh_noise_robustness_L2_fix1B_
    heavy_phase2d_all_noisy_strict.json --output-pdf /Users/jakestrobel/Documents/Holstein_implementation/Hols
    tein_test/docs/HH_noise_robustness_L2_fix1B_heavy_phase2d_all_noisy_strict.pdf
```